

MANEJO DE PACIENTES CON RIESGO DE MRONJ

Basado en la AAOMS Position Paper 2022 (se adjunta artículo)

1. Definición de MRONJ

La Osteonecrosis de los Maxilares relacionada con Medicamentos (en inglés MRONJ) se diagnostica si se cumplen **todos** los siguientes criterios:

1. Exposición actual o previa a fármacos antiresortivos (bisfosfonatos, denosumab, romosozumab...) y/o antiangiogénicos/inmunomoduladores.
2. Hueso necrótico expuesto o sondable a través de fístula intra/extraoral en región maxilofacial o cervicofacial, presente >8 semanas de evolución.
3. Ausencia de radioterapia previa o metástasis en los maxilares.

2. Fármacos implicados

Bisfosfonatos: Fármacos antiresortivos que inhiben la actividad osteoclástica, reduciendo la resorción ósea. Se usan para tratamiento de osteoporosis, metástasis óseas y otras patologías.

Se dividen en:

VÍA DE ADMINISTRACIÓN	EJEMPLOS	RIESGO MRONJ
ORAL	Alendronato, Risedronato, Ibandronato	Bajo (<0,05%)
INTRAVENOSA	Ácido zoledrónico, Pamidronato	Alto (<5% en cáncer, hasta 18% en series de casos extensas)

Denosumab: Anticuerpo monoclonal que inhibe RANKL, bloqueando la formación y función de osteoclastos. Se administra por vía subcutánea. Riesgo MRONJ: 0,3% en osteoporosis; <7% en oncología.

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3. Factores de riesgo

Tipo de factor	Detalle
Fármaco	Bisfosfonatos IV y denosumab en oncología: alto riesgo
Duración	>2 años (más marcado en oncología)
Locales	Exodoncias, implantes, prótesis mal ajustadas, enfermedad periodontal
Sistémicos	Cáncer, corticoides, quimioterapia, diabetes, tabaco
Anatómicos	Mandíbula (75%) > Maxila (25%)

4. Clasificación AAOMS 2022

Estadio	Descripción
En riesgo	Sin hueso necrótico, asintomático, tratamiento de riesgo
0	Sin hueso necrótico expuesto, síntomas inespecíficos o hallazgos radiográficos
1	Hueso necrótico expuesto/sondable, asintomático, sin infección
2	Igual que 1 + infección y dolor
3	Igual que 2 + complicaciones (fractura, fístulas, osteólisis extensa)

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5. Hallazgos radiográficos sospechosos de MRONJ *

*Bedogni A, Mauceri R, Fusco V, Bertoldo F, Bettini G, Di Fede O, Lo Casto A, Marchetti C, Panzarella V, Saia G, Vescovi P, Campisi G. Italian position paper (SIPMO-SICMF) on medication-related osteonecrosis of the jaw (MRONJ). Oral Dis. 2024 Sep;30(6):3679-3709. doi: 10.1111/odi.14887. Epub 2024 Feb 5. PMID: 38317291.

Radiological signs of MRONJ	Plain radiographs	CT-based radiological investigations
Cortical erosion	X	✓
Diffuse bone marrow sclerosis ^a	✓	✓
Focal bone marrow sclerosis ^a	X	✓
Opacified maxillary sinus	✓	✓
Osteolytic changes	X	✓
Osteolysis extending to the maxillary sinus	X	✓
Osteosclerosis of adjacent bones (zygoma and hard palate)	X	✓
Pathologic fracture	✓	✓
Periodontal space widening	✓	✓
Periosteal reaction	✓	✓
Persistent post-extraction socket	✓	✓
Sequester formation	✓	✓
Sinus tract ^b	X	✓
Thickening of the alveolar ridge	✓	✓
Thickening of the lamina dura	✓	✓
Thickening of the inferior alveolar nerve canal	✓	✓
Trabecular thickening	X	✓

Note: ✓, detectable. X, undetectable or detectable only in advanced MRONJ cases.

^aSclerosis: trabecular bone disorganization and poor corticomedullary differentiation.

^bOroantral, oronasal or orocutaneous communication due to bone destruction.

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6. Algoritmo de prevención antes de iniciar tratamiento con antiresortivos

1. Evaluar salud oral (clínica y radiográfica).
2. Eliminar focos infecciosos y dientes con mal pronóstico.
3. Ajustar prótesis para evitar ulceraciones.
4. Educar sobre higiene oral y riesgos.
5. Iniciar medicación solo cuando mucosa esté cicatrizada post-cirugía.

7. Algoritmo de manejo según estadio

Estadio	Manejo
En riesgo / 0	Educación, higiene oral, control periódico
1	Clorhexidina 0,12-0,2%, cada 12 h / 15 días + vigilancia
2	Igual que 1 + antibióticos si infección
3	Igual que 2 + cirugía por cirujanos maxilofaciales si indicado (secuestrectomías, fracturas mandibulares...)

Nota: ver “*Consenso SECOM CyC*” adjuntado al protocolo (pág. 6-8)

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8. Algoritmo de actuación en tratamientos dentales en pacientes a tratamiento con bisfosfonatos

PROCEDIMIENTO	BF y VO < 4 años	BF y VO > 4 años	BF y VI ONCOLÓGICO
EXODONCIA	Llevar a cabo el tratamiento	Suspender BF 2-3 meses*	No realizar tratamiento ¹
CIRUGÍA DE IMPLANTES Y CIRUGIA PERIODONTAL	Llevar a cabo el tratamiento	Suspender BF 2-3 meses*	No realizar tratamiento
PERIODONCIA (PROFILAXIS Y RAR)	Llevar a cabo el tratamiento	Llevar a cabo el tratamiento	Llevar a cabo el tratamiento
ENDODONCIA	Llevar a cabo el tratamiento	Llevar a cabo el tratamiento	Llevar a cabo el tratamiento
PRÓTESIS	Llevar a cabo el tratamiento	Llevar a cabo el tratamiento	Llevar a cabo el tratamiento
ORTODONCIA	Viable pero no recomendado	Viable pero no recomendado	No realizar tratamiento

BF= Bisfosfonato / VO = vía oral / VI = vía intravenosa por motivos oncológicos / RAR= raspado y alisado radicular

*Suspender el tratamiento 2-3 meses, realizar el procedimiento y esperar 3 meses hasta reintroducir el BF.

¹Solo realizar en caso de necesidad cuando fallaron las medidas conservadoras (endodoncia o endodoncia + decapitación corona clínica)

Black DM, Reid IR, Boonen S, et al: The effect of 3 versus 6 years of zoledronic acid treatment of osteoporosis: A randomized extension to the HORIZON-Pivotal fracture trial (PFT). J Bone Miner Res 27:243, 2012

Black DM, Reid IR, Cauley JA, et al: The effect of 6 versus 9 years of zoledronic acid treatment in osteoporosis: A randomized second extension to the HORIZON-Pivotal fracture trial (PFT). J Bone Miner Res 30:934, 2015

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9. Algoritmo de actuación en tratamientos dentales en pacientes a tratamiento con DENOSUMAB (Prolia - no oncológico // Xgeva - oncológico)

PROCEDIMIENTO	DENOSUMAB (Prolia®)
EXODONCIA	Llevar a cabo el tratamiento*
CIRUGÍA DE IMPLANTES Y CIRUGIA PERIODONTAL	Llevar a cabo el tratamiento*
PERIODONCIA (PROFILAXIS Y RAR)	Llevar a cabo el tratamiento
ENDODONCIA	Llevar a cabo el tratamiento
PRÓTESIS	Llevar a cabo el tratamiento
ORTODONCIA	Viable pero no recomendado

RAR= raspado y alisado radicular

*suspender el tratamiento o esperar 5 los meses después de la última administración del denosumab y colocar la siguiente dosis una vez que se haya objetivado la curación de la zona intervenida (4-6 semanas aproximadamente tras la cirugía)

¹Bedogni A, Mauceri R, Fusco V, Bertoldo F, Bettini G, Di Fede O, Lo Casto A, Marchetti C, Panzarella V, Saia G, Vescovi P, Campisi G. Italian position paper (SIPMO-SICMF) on medication-related osteonecrosis of the jaw (MRONJ). Oral Dis. 2024 Sep;30(6):3679-3709. doi: 10.1111/odi.14887. Epub 2024 Feb 5. PMID: 38317291.

PROCEDIMIENTO	DENOSUMAB (Xgeva®)
EXODONCIA	Llevar a cabo el tratamiento*
CIRUGÍA DE IMPLANTES Y CIRUGIA PERIODONTAL	No realizar tratamiento
PERIODONCIA (PROFILAXIS Y RAR)	Llevar a cabo el tratamiento
ENDODONCIA	Llevar a cabo el tratamiento
PRÓTESIS	Llevar a cabo el tratamiento
ORTODONCIA	Viable pero no recomendado

RAR= raspado y alisado radicular

* esperar hasta la 3ª semana de la última dosis del DMB y colocar la siguiente dosis una vez que se haya objetivado la curación de la zona intervenida (4-6 semanas aproximadamente tras la cirugía). Si las condiciones sistémicas y el médico prescriptor lo permiten.

Bedogni A, Mauceri R, Fusco V, Bertoldo F, Bettini G, Di Fede O, Lo Casto A, Marchetti C, Panzarella V, Saia G, Vescovi P, Campisi G. Italian position paper (SIPMO-SICMF) on medication-related osteonecrosis of the jaw (MRONJ). Oral Dis. 2024 Sep;30(6):3679-3709. doi: 10.1111/odi.14887. Epub 2024 Feb 5. PMID: 38317291.

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EVALUACION DE UNA MRONJ

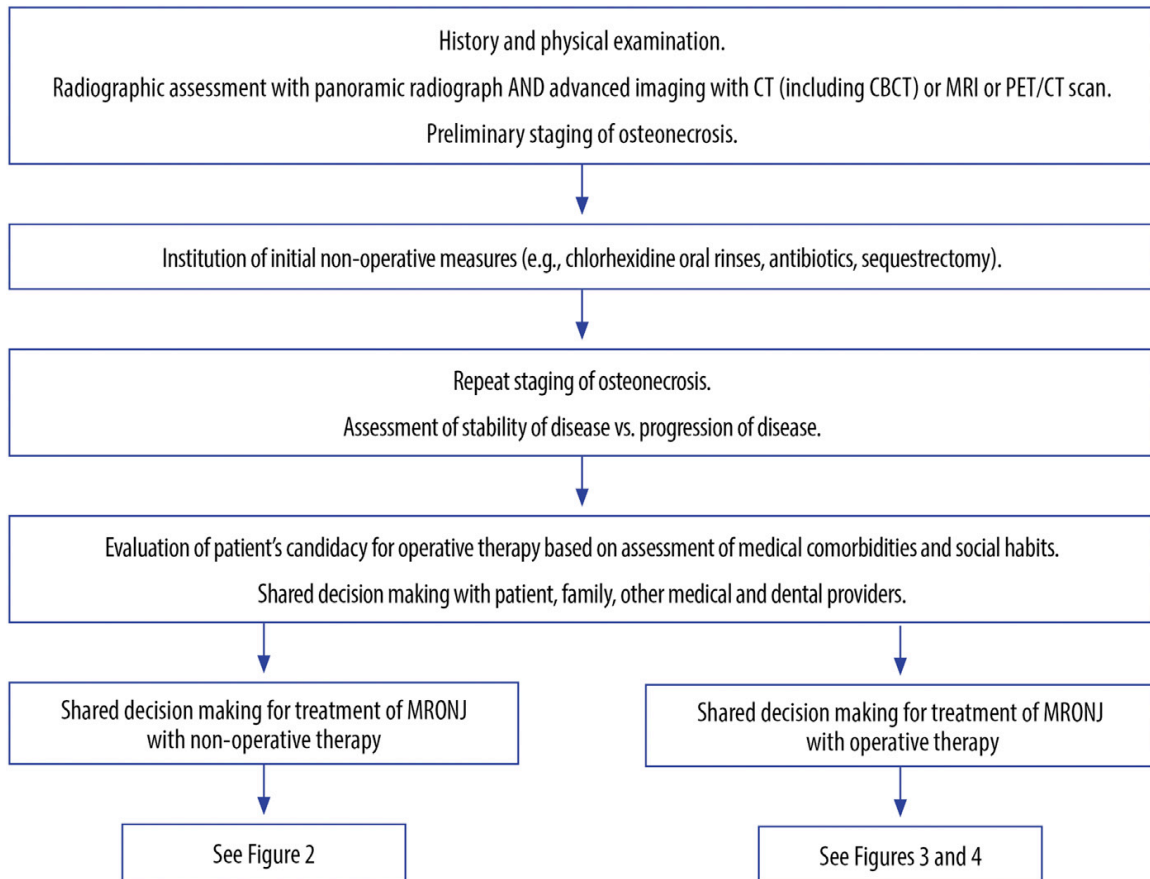
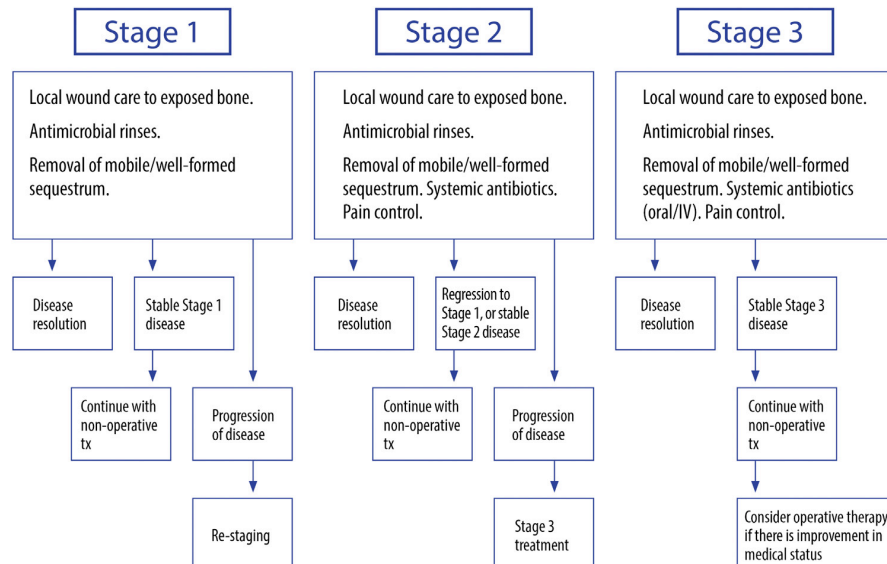


FIGURE 2



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FIGURE 3

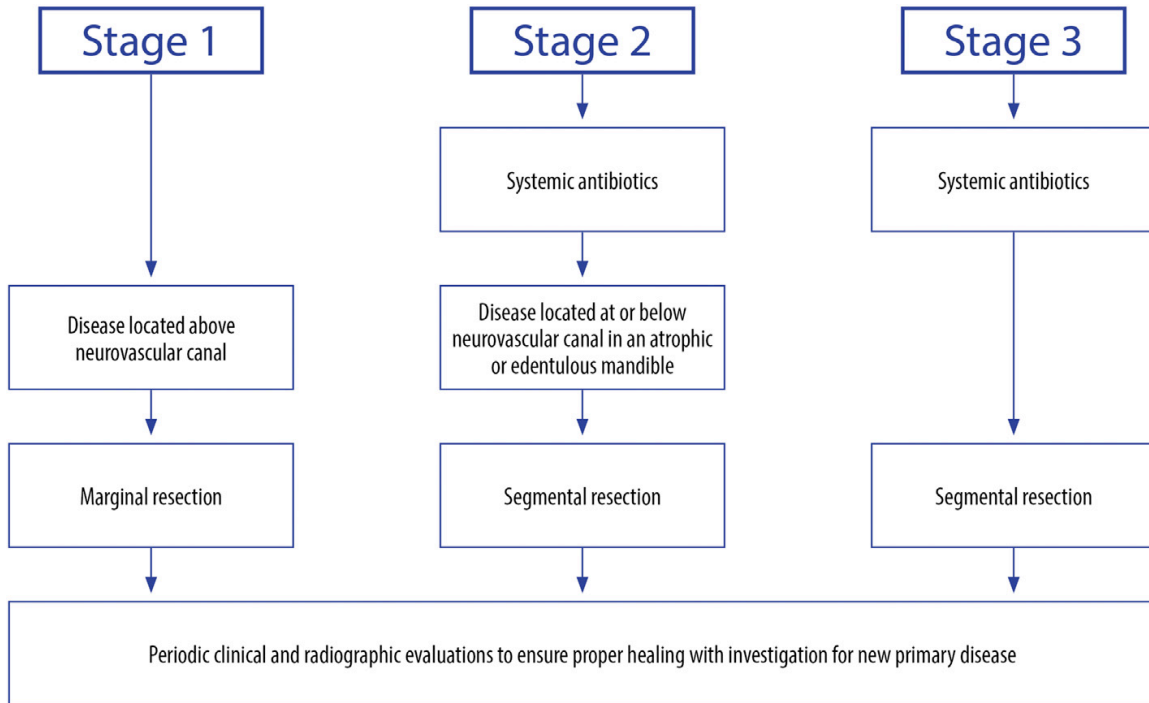
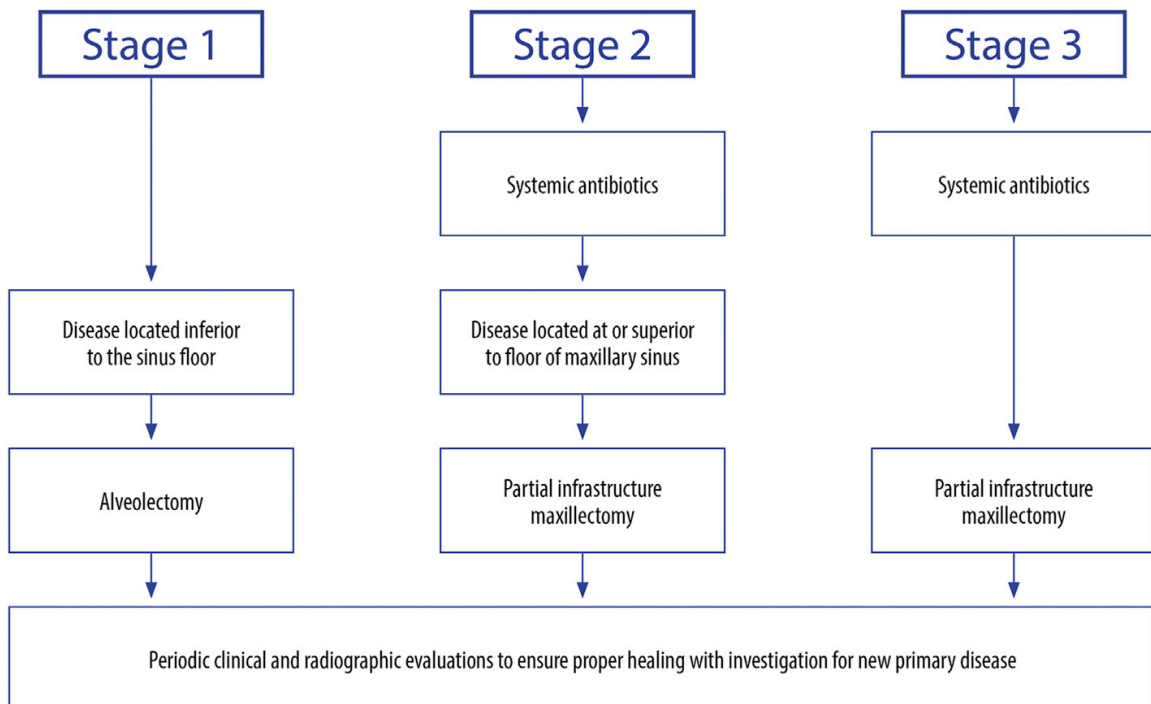


FIGURE 4



American Association of Oral and Maxillofacial Surgeons' Position Paper on Medication-Related Osteonecrosis of the Jaws—2022 Update



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Strategies for management of patients with, or at risk for, medication-related osteonecrosis of the jaws (MRONJ) – formerly referred to as bisphosphonate-related osteonecrosis of the jaws (BRONJ)—were set forth in the American Association of Oral and Maxillofacial Surgeons (AAOMS) position papers in 2007, 2009 and 2014. The position papers were developed by a committee appointed by the AAOMS Board of Trustees and comprising clinicians with extensive experience in caring for these patients, as well as clinical and basic science researchers. The knowledge base and experience in addressing MRONJ continues to evolve and expand, necessitating modifications and refinements to the previous position papers. Three members of the AAOMS Committee on Oral, Head, and Neck Oncologic and Reconstructive Surgery (COH-NORS) and three authors of the 2014 position paper were appointed to serve as a working group to analyze the current literature and revise the guidance as indicated to reflect current knowledge in this field. This update contains revisions to diagnosis and management strategies and highlights the current research status. AAOMS maintains that it is vitally important for this information to be disseminated to other relevant healthcare professionals and organizations.

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Medications prescribed for dental and medical conditions have potential side effects that warrant a risk-benefit discussion. Where therapeutic margins are

wide and complications are readily corrected, decisions are implemented in a straightforward fashion. Where therapeutic margins are wide but

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complications are significant, deciding to proceed with pharmacologic treatment becomes more challenging. In most cases of MRONJ, local therapies can be successful. The fact that more complex treatment is required for a few patients should not impact decision-making for all other patients with osteonecrosis of the jaws. The medications associated with MRONJ have proved to be safe and effective in clinical trials and postmarketing analyses for most patients and should continue as a mainstay therapy when indicated. Communicating the risks of MRONJ to patients and providers is critical to ensure appropriate medical management for the primary disease.

Undoubtedly, risk profiles may change as new medications come to market. In addition, our understanding of disease pathophysiology, risk modifiers, and treatment strategies will continue to evolve. It is of the utmost importance that clinicians base their patient treatment decisions on currently available scientific evidence.

Strategies for management of patients at risk for or with MRONJ were set forth in American Association of Oral and Maxillofacial Surgeons (AAOMS) Position Papers in 2007,¹ 2009,² and 2014.³ These position papers were developed by a committee appointed by the AAOMS Board of Trustees and comprised of clinicians with extensive experience in caring for these patients, as well as clinical and basic science researchers. The knowledge base and experience in addressing MRONJ continues to evolve and expand, necessitating modifications and refinements to the previously published position papers. A working group comprised of three members of the AAOMS Committee on Oral, Head, and Neck Oncologic and Reconstructive Surgery (COH-NORS) and three authors of the 2014 paper convened remotely in the fall of 2020 to appraise the current literature and revise the guidelines as indicated to reflect the current knowledge in this field. This update contains revisions to the pathogenesis and management strategies and highlights the current research status. AAOMS maintains it is vitally important for this information to be disseminated to other relevant healthcare professionals and organizations.

Purpose

The purpose of this position paper is to provide updates regarding:

1. Risk estimates for developing MRONJ.
2. Comparisons of the risks and benefits of medications related to osteonecrosis of the jaw in order to facilitate medical decision-making for the treating physician, dentist, dental specialist, and patient with the establishment of algorithms.

3. Guidance to clinicians regarding:
 - a. The differential diagnosis of MRONJ in patients with a history of exposure to antiresorptive medications.
 - b. MRONJ prevention measures and management strategies for patients with MRONJ based on the disease stage.

Medications

Bisphosphonates (BPs) are antiresorptive medications that are effective in managing cancer-related conditions, including hypercalcemia of malignancy, spinal cord compression, and pathologic fractures (skeletal-related events [SREs]) associated with bone metastases in the context of solid tumors (such as breast, prostate, and lung cancers) and multiple myeloma.⁴⁻¹³ While the potential for BPs to improve cancer-specific survival remains controversial, these medications have had a significant positive effect on the quality of life for patients with advanced cancer involving the skeleton and reducing or preventing skeletal-related events.

Bisphosphonates also are used for the prevention of osteoporosis-related fractures (fragility fractures) in patients with osteoporosis and osteopenia.¹⁴⁻¹⁶ BPs administered orally—including alendronate (Fosamax®), risedronate (Actonel®) or parenterally (zoledronic acid [Reclast®]), and ibandronate (Boniva®)—can result in a significant reduction in vertebral and nonvertebral fractures for patients with osteoporosis.¹⁷⁻²⁰

Bisphosphonate therapy also is indicated for other metabolic bone diseases such as Paget's disease of bone and osteogenesis imperfecta.²¹⁻²³ However, clinical trials have not demonstrated the efficacy of bisphosphonate therapy in the management of fibrous dysplasia.²⁴

Denosumab (DMB), a receptor activator of nuclear factor kappa-B ligand (RANK-L), is an antiresorptive agent that exists as a fully humanized antibody against RANK ligand and inhibits osteoclast function and associated bone resorption. When denosumab (Prolia®) is administered subcutaneously every 6 months, there is a significant reduction in the risk of vertebral, nonvertebral, and hip fractures in osteoporotic patients.²⁵⁻²⁸ Denosumab (Xgeva®) also is effective in reducing SREs related to metastatic bone disease from solid tumors when administered monthly.²⁹⁻³¹

RANK ligand inhibitors also have proven efficacy in the treatment of giant cell tumors of bone and fibrous dysplasia.³²⁻³⁶ In contrast to BPs, RANK-L inhibitors do not bind to bone, and their effects on bone remodeling are mostly diminished within 6 months of treatment cessation.

Romosozumab is a new monoclonal antibody used for fracture prevention in osteoporotic women. Romosozumab, administered subcutaneously, works via the Wnt pathway by binding to and inhibiting sclerostin, resulting in increased bone formation and decreased bone resorption.³⁷

MRONJ Case Definition

MRONJ should be distinguished from other forms of osteonecrosis (ONJ) conditions and identified by history and clinical exam. The clinical criteria required to establish a diagnosis of MRONJ have remained unchanged from the previous position paper.³

The case definition of MRONJ includes all the following elements:

1. Current or previous treatment with antiresorptive therapy alone or in combination with immune modulators or antiangiogenic medications.
2. Exposed bone or bone that can be probed through an intraoral or extraoral fistula(e) in the maxillofacial region that has persisted for more than 8 weeks.
3. No history of radiation therapy to the jaws or metastatic disease to the jaws.

Staging

A staging system for MRONJ was introduced in the 2009 AAOMS position paper and then modified in the 2014 position paper to characterize more accurately all aspects of the clinical presentation of MRONJ. Since these modifications, the AAOMS staging system has continued to be a straightforward and relevant system to properly stratify these patients. It has been adopted by several professional societies and research organizations. The staging system facilitates the creation of rational treatment guidelines and guides data collection to assess the prognosis and outcomes for MRONJ patients. While AAOMS recognizes that different classification systems are being used by other organizations,³⁸ the Association considers the AAOMS system to be a useful and widely implemented assessment tool guiding clinicians involved in the care of MRONJ patients. AAOMS remains concerned that overemphasizing variable radiographic features often attributed to MRONJ may overestimate the true disease frequency by including false positives in the numerator (eg, cases with radiographic findings suggestive of MRONJ), but these patients do not fit the criteria for the diagnosis of MRONJ. In the orthopedic literature, the usefulness of a Stage 0 category has been established for staging avascular necrosis (AVN) of the femoral head when there is a suspicion of AVN in a pa-

tient at risk, but the diagnostic information is not conclusive.³⁹ AAOMS believes the Stage 0 category for MRONJ is analogous in principle and does account for the wide-ranging radiographic presentation of MRONJ that exists prior to overt bone exposure. Therefore, AAOMS has decided to maintain the current classification system with no modifications.

PATIENTS AT-RISK

No apparent necrotic bone in asymptomatic patients who have been treated with IV or oral antiresorptive therapy.

STAGE 0 (NONEXPOSED BONE VARIANT)

Patients with no clinical evidence of necrotic bone but who present with nonspecific symptoms or clinical *and* radiographic findings, such as:

Symptoms

- Odontalgia not explained by an odontogenic cause.
- Dull, aching bone pain in the jaw, which may radiate to the temporomandibular joint region.
- Sinus pain, which may be associated with inflammation and thickening of the maxillary sinus wall.
- Altered neurosensory function.

Clinical Findings

- Loosening of teeth not explained by chronic periodontal disease.
- Intraoral or extraoral swelling.

Radiographic Findings

- Alveolar bone loss or resorption not attributable to chronic periodontal disease.
- Changes to trabecular pattern sclerotic bone and no new bone in extraction sockets.
- Regions of osteosclerosis involving the alveolar bone and/or the surrounding basilar bone.
- Thickening/obscuring of periodontal ligament (thickening of the lamina dura, sclerosis, and decreased size of the periodontal ligament space).⁴⁰

These nonspecific findings, which characterize this variant of MRONJ without bone exposure, may occur in patients with a prior history of Stage 1, 2, or 3 disease who have been healed and have no clinical evidence of exposed bone. Progression to Stage 1 disease has been reported in up to 50 percent of

patients with Stage 0 disease⁴¹ and, therefore, AAOMS deems it prudent to consider Stage 0 disease as a potential precursor to MRONJ.

STAGE 1

Exposed and necrotic bone or fistula that probes to the bone in patients who are asymptomatic and have no evidence of infection/inflammation. These patients also may present with radiographic findings mentioned for Stage 0 that are localized to the alveolar bone region.

STAGE 2

Exposed and necrotic bone, or fistula that probes to the bone, with evidence of infection/inflammation. These patients are symptomatic. These patients also may present with radiographic findings mentioned for Stage 0 localized to the alveolar bone region.

STAGE 3

Exposed and necrotic bone or fistulae that probes to the bone, with evidence of infection, and one or more of the following:

- Exposed necrotic bone extending beyond the region of alveolar bone (ie, inferior border and ramus in the mandible, maxillary sinus, and zygoma in the maxilla)
- Pathologic fracture.
- Extraoral fistula.
- Oral antral/oral-nasal communication.
- Osteolysis extending to the inferior border of the mandible or sinus floor.

CAUSALITY

It is important to understand that patients at risk for or with established MRONJ also can present with other common clinical conditions not to be confused with MRONJ. Commonly misdiagnosed conditions may include but are not limited to alveolar osteitis, sinusitis, gingivitis/periodontitis, caries, periapical pathology, odontalgia, atypical neuralgias, fibro-osseous lesions, sarcoma, chronic sclerosing osteomyelitis, and temporomandibular joint (TMJ) disorders. It also is important to remember that delayed healing, exposed bone or sequestra (ie, osteonecrosis [ONJ]), can occur in patients not exposed to antiresorptive agents.⁴²

Proving causality of any medication-related complication is challenging from an epidemiologic perspective. It is well-known that MRONJ is a rare entity, multifactorial in nature, and patients with the same clinical presentation exist who have not been exposed to an antiresorptive medication. Studies have reported

jaw necrosis in antiresorptive naïve patients in which necrosis was linked to bacterial, viral, or fungal infections, trauma, smoking, steroids, immunocompromised host, autoimmune diseases, diabetes, and chemotherapy.⁴³⁻⁵⁷ Furthermore, patient populations, such as those with osteogenesis imperfecta have been treated with antiresorptive agents without reports of MRONJ.⁵⁸ Many patients receiving medications associated with MRONJ have other comorbidities, which are likely exacerbating or contributing factors. In combination, these confounding variables make incidence and prevalence difficult to estimate.

Clinical trials, while being the gold standard for efficacy and safety data, are seldom powered to demonstrate uncommon events. Prior to the discovery of MRONJ, large randomized prospective trials of BPs with up to 10 years of patient data did not reveal any jaw bone necrosis as a complication.^{17,59} More recently, the HORIZON Pivotal Fracture trial tested 3,889 randomized patients given annual zoledronic acid versus placebo for 3 years; one patient developed MRONJ in the intervention group and one in the placebo group.¹⁸ Extension of this trial for up to 6 years resulted in one additional MRONJ patient in the treatment group.⁶⁰ Extension to 9 years resulted in no additional confirmed cases of MRONJ.⁶¹

Definitive causality, taken as a whole, remains a difficult task to prove in general, let alone in individual patients presenting with clinical symptoms. Clinicians should be aware of these facts in decisions regarding treatment recommendations.

PATHOPHYSIOLOGY

Since the AAOMS position paper in 2014, significant knowledge has been gained regarding MRONJ pathophysiology from both clinical and particularly preclinical animal studies. It should be noted that animal studies have a number of limitations, are most often using supratherapeutic doses and likely do not truly mirror the clinical environment. That said, they are critical in understanding disease mechanisms and can serve as one reference point to evidence-based clinical decision-making.

Much debate persists among clinicians and researchers, contributing to the various treatment protocols utilized for patients today.⁶²⁻⁶⁵ Disease specificity unique to the jaws has focused leading hypotheses to include bone remodeling inhibition, inflammation or infection, angiogenesis inhibition, innate or acquired immune dysfunction, as well as genetic predisposition.^{3,65} Both animal and human studies suggest that an antiresorptive medication, coupled with inflammation or infection, is necessary and

sufficient to induce MRONJ. However, as more knowledge is gained on the subject, it is becoming increasingly apparent that MRONJ is multifactorial, and it is likely that multiple hypotheses can explain the overall pathophysiology of this disease.^{3,65}

Bone Remodeling Inhibition

The definition of MRONJ includes oral or parenteral administration of antiresorptive medications, such that bone remodeling suppression is a central hypothesis in its pathophysiology. Antiresorptive medications, including BPs and denosumab (DMB), have direct effects on osteoclast formation, differentiation, or function. In osteoporosis, BPs are a first-line therapy to decrease bone remodeling, increase bone mineral density, and decrease vertebral and long bone fractures.^{66,67} BPs, in higher doses, also are utilized in primary bone malignancy and bone metastases to decrease SREs, including hypercalcemia of malignancy, reduce severe bone pain, and improve quality of life.⁶⁸⁻⁷² Although DMB has only been approved for use since 2010, its use has increased significantly for both osteoporosis and malignancy in the last decade. Prevalence of MRONJ with DMB users is at least as high as BP users, likely due to its increased potency to inhibit bone resorption.^{30,73-75} This is supported in the jaws as animal studies demonstrate absent osteoclasts around the alveolar bone of DMB-treated mice.⁷⁶ Human bone specimens also show an increased number of nonfunctional osteoclasts surrounding necrotic bone in BP-treated patients,⁷⁷ further reinforcing bone remodeling inhibition as a leading hypothesis in MRONJ pathophysiology. With the appearance of MRONJ in DMB-treated patients, it becomes increasingly apparent that the underlying pathophysiology involves dysfunctional osteoclasts.

Animal studies evaluating withdrawal of BPs or DMB further highlight the importance of bone remodeling in MRONJ prevention and resolution. Rodents with established ONJ failed to resolve when antiresorptive were withdrawn. However, discontinuing DMB, but not BPs, prior to tooth extraction successfully prevented MRONJ development in rats.^{78,79} Moreover, parathyroid hormone, which acts directly on osteoblasts to induce bone formation and indirectly increases osteoclastic bone resorption and overall remodeling, has been shown to prevent MRONJ and improve extraction socket healing in rodents and preliminarily in patients.⁸⁰⁻⁸² This observation provides further support for the central role of osteoclast inhibition in MRONJ pathogenesis.

Inflammation or Infection

Although most studies report tooth extraction as the major inciting event for MRONJ development, it is clear that most extracted teeth had pre-existing peri-

odontal or periapical disease.^{3,64,83,84} From this patient information, animal models of inflammation or infection were developed to replicate clinical, radiographic, and histologic features of MRONJ.⁸⁵⁻⁸⁸ Presence of inflammatory cytokines, specifically at the site of MRONJ, also support the strong role of inflammation.⁸⁹ As evidence of increased systemic inflammation and its contribution to MRONJ development, mice with experimentally induced rheumatoid arthritis demonstrated more severe MRONJ with increased oral bone exposure, more pronounced radiographic features, intense local inflammatory infiltrate, and larger areas of histologic necrosis.⁹⁰ Further support for the inflammatory etiology showed that removal of the inflammatory nidus in ligature-induced periodontitis ameliorated MRONJ development in mice, demonstrating reduced inflammation and prevention of disease progression.⁹¹ Moreover, transplantation of peripheral blood mononuclear cells with anti-inflammatory properties reduced MRONJ prevalence by improving soft-tissue healing, decreasing inflammatory polymorphonuclear cells and inflammatory marker expression, as well as enhancing vascularity.⁹² These preclinical findings confirm the irrefutable role of inflammation or infection in MRONJ disease prevalence, severity and resolution.

The presence of bacteria on the exposed necrotic bone also contributes to disease severity, where pain and signs of infection define Stage 2 MRONJ.^{3,93,94} This is not surprising since poor oral hygiene and biofilm presence are associated with MRONJ development,^{95,96} and oral health maintenance and dental prophylaxis before initiating antiresorptive therapy can decrease MRONJ prevalence.^{97,98} Importantly, clinical treatment protocols to reduce the biofilm and eradicate infection have emerged as important alternatives to debridement and resection in patients who may not be ideal surgical candidates.⁶³

Angiogenesis Inhibition

Osteonecrosis is traditionally defined as avascular necrosis or aseptic necrosis, most commonly characterized as osteocyte death after decreased blood flow to the femoral head.⁹⁹ However, MRONJ is defined as necrotic bone in the maxillofacial region after exposure to either antiresorptive or antiangiogenic medications.³ BPs such as zoledronic acid directly inhibit angiogenesis in vitro and in vivo¹⁰⁰⁻¹⁰³ and animal models demonstrate decreased vascularity in sites of MRONJ and decreased microvessel numbers during early stages of bone healing.¹⁰⁴ In addition, angiogenesis normally seen during extraction socket healing is inhibited by BPs, and both BPs and DMB have been shown to decrease arterial area, venous area, and overall vascularity of periodontal tissues during

early and late MRONJ development.^{105,106} Importantly, antiangiogenic medications, such as VEGF inhibitors, tyrosine kinase receptor inhibitors, and immunomodulatory drugs,¹⁰⁷⁻¹⁰⁹ can be associated with MRONJ. Moreover, patients with multiple myeloma receiving both antiresorptive and antiangiogenic medications, as shown in several studies,¹¹⁰⁻¹¹⁴ have a higher MRONJ prevalence. Important aspects of MRONJ treatment include determining disease margins, which can be challenging as microvascular mucosal abnormalities can be seen adjacent to frank MRONJ lesions.¹¹⁵ It is important to note that the incidence of MRONJ in patients on antiangiogenics is much lower than those taking antiresorptive medications.

Innate or Acquired Immune Dysfunction

Although animal studies confirm that an antiresorptive medication—coupled with inflammation or infection—is necessary and sufficient to produce MRONJ, not all patients with dental infections develop the disease. It is well-known that patients with medical comorbidities such as diabetes or rheumatoid arthritis or immunocompromised states are at significantly higher risk for MRONJ with or without exposure to antiresorptive agents.^{3,64,114,116} Patients with metastatic or primary bone malignancies have a compromised immune system.¹¹⁷ This also has been confirmed with animal studies, where chemotherapy, steroids, and disease-modifying antirheumatic drugs (DMARDs), combined with antiangiogenic medications and an antiresorptive, increase MRONJ severity or prevalence.¹¹⁸⁻¹²⁰ Moreover, higher rates of MRONJ occur in patients with multiple myeloma who receive multiple chemotherapeutic agents.^{110,121}

Replenishing the area of nonhealing MRONJ lesions with mesenchymal stem cells (MSCs) to overcome immune dysfunction is a potential area of therapeutic interest, especially in patients who are immunocompromised. A recent study showed altered numbers and patterns of T-cells in human and rat MRONJ necrotic bone samples as compared to healthy patients and non-MRONJ sites.¹²² Preclinical studies also demonstrate healing or prevention of MRONJ lesions after systemic infusion with adipose or bone marrow-derived MSCs.¹²³⁻¹²⁵

Genetic Factors

In the 2014 paper, the authors identified several reports describing single-nucleotide polymorphisms (SNPs) that were associated with the development of MRONJ. Most of these SNPs were located within regions of the gene associated with either bone turnover, collagen formation, or certain metabolic bone diseases. Indeed, increasing evidence is available to support the role of SNPs with MRONJ.^{126,127} Specific

links to sirtuin-1 (SIRT1), a bone remodeling regulator that promotes bone formation, may be protective against MRONJ if upregulated.¹²⁸ SIRT1 also is involved in both reduction of inflammation and induction of angiogenesis, suggesting a role in several of the leading MRONJ hypotheses.¹²⁸ Other genes also have been reported to increase MRONJ risk through their role in angiogenesis, bone remodeling, and immune responses, including PPAR gamma, CYP2C8, and many others.¹²⁹ Collectively, these studies suggest that MRONJ is a multifactorial disease and that genetic factors may play a role in its development.¹³⁰ Overall, however, current studies document either a weak or no association between genetic factors measured and risk for MRONJ.¹³¹ To determine predisposition, studies with larger sample sizes should be performed, with genetic risks confirmed in both BPs and DMB-treated patients who have breast or prostate cancer metastases, multiple myeloma, or osteoporosis.

Risk Factors for MRONJ

MEDICATION-RELATED RISK FACTORS

To estimate the risk for medications associated with MRONJ, the primary parameter to be considered is the therapeutic indication for treatment (eg, malignancy or osteoporosis/osteopenia). The data suggest that antiresorptive medications (ie, BPs and DMB) are associated with an increased risk for developing MRONJ. The risk of MRONJ is considerably higher in the malignancy group (<5%) than in the osteoporosis group (<0.05%). Current data are insufficient to identify other medications as risk factors for developing MRONJ.

MRONJ risk among cancer patients

For estimating the risk for MRONJ among patients exposed to a medication, the risk for MRONJ in patients *not exposed* to antiresorptive medications must be estimated (Table 1). The risk for MRONJ among cancer patients enrolled in clinical trials and assigned to *placebo* groups ranges from 0 percent to 0.7 percent.¹³²⁻¹³⁸

- a. Among cancer patients exposed to zoledronate, the cumulative risk of MRONJ clusters in the low single digits, <5 percent, and ranges from 0 percent to 18 percent.^{113,132,133,137-144} The wide variation in estimates may be explained by the varying durations of follow-up, one to 10 years, reported in the various studies. The risk of MRONJ among cancer patients exposed to zoledronate ranges between 2-10 times higher than cancer patients treated with placebo.
- b. Among cancer patients exposed to DMB, the risk of MRONJ ranges from 0 percent to 6.9 percent, with most studies reporting rates <5

Table 1. MRONJ DISEASE FREQUENCY GROUPED BY INDICATIONS FOR TREATMENT VERSUS MEDICATION*

Indications for Treatment	Medication					Study Design
	Placebo	Zoledronate	Oral BPs	Denosumab	Romosozumab	
Malignancy						
Coleman (2020) ¹³⁸	0.2% (2,218)*			5% (2,241)		RCT [†]
O’Carrigan et al, (2017) ¹³⁷	0.7% (6,788)			0.4% (6,788)		Systematic review
O’Carrigan et al, (2017) ¹³⁷	0% (3,060)	1% (3,078)				Systematic review
Macherey et al, (2017) ¹³⁶	0.7% (818)	1.5% (808)				Systematic review
Gnant et al, (2015) ²⁴⁷	0% (903)	0% (900)				RCT
Coleman et al, (2014) ¹³³	0% (1,679)	1.7% (1,681)				RCT
Valachis et al, (2013) ¹³²	0% (3,039)	0.52% (4,774)				Systemic review
Boquete-Castro et al, (2016) ¹³⁵	0.1%	1.14%		1.7%		Systematic review
Coleman (2020) ¹³⁸	0.2% (2,218)			5.4% (2,214)		RCT
Gnant et al, (2015) ²⁴⁷	0% (1,709)			0% (1,711)		RCT
Raje et al, (2018) ¹¹³		2.8% (82)		4.1% (850)		RCT
Himelstein (2017) ¹⁴⁰		1.5% (1,822)				RCT
Henry (2014) ¹⁴¹		1.1% (786)		0.8% (792)		RCT
Yang et al, (2019) ²⁴⁸		2% (8,525)				Systematic review
Peddi et al, (2013) ¹⁴²		1.3% (2,846)		1.8% (2,885)		Systematic review
Ng et al, (2021) ¹⁴⁵		1.6-4% [‡] 3.8-18% [§]		1.9% [‡] 6.9% [§]		Systematic review
Wang et al, (2014) ¹⁴⁴		1.4% (1,013)		2% (1,020)		Systematic review
Osteoporosis						
Papapoulos et al, (2012) ²⁶	0% (3,383)			0.04% (4,549)		RCT
Grbic et al, (2010) ¹⁵⁰	0.02% (4,945)	0.02% (5,864)				Systematic review
Cosman et al, (2016) ¹⁵¹	0% (3,322)				0.03% (3,321)	RCT
Saag et al, (2017) ³⁷			0.05% (2,047)		0.05% (2,046)	RCT
Bone et al, (2017) ¹⁵³				0.3% (2,343) 10-yr f/u		RCT
Hallmer et al, (2018) ⁷⁵			0.043%			Population study (50,000)
Nonmalignant bone disease						
Chawla et al, (2019) ¹⁵⁶				5% (532)		Prospective case series
Rutkowski ¹⁵⁵				0.7% (138)		Retrospective case series

* Sample size in parentheses

† Randomized clinical trial.

‡ <2 years of follow-up.

§ >2 years of follow-up.

percent.^{113,134,135,138,141,142,144,145} The risk for MRONJ among cancer patients exposed to DMB is comparable to the risk of MRONJ in cancer patients exposed to zoledronate.^{135,141,142,144,145}

Since the 2014 update, investigators have implicated numerous families of medications as risk factors for MRONJ.¹⁴⁶⁻¹⁴⁹ These medications include tyrosine kinase inhibitors (TKIs) such as sunitinib, monoclonal antibodies (bevacizumab), fusion proteins (afibercept), mTOR inhibitors (everolimus), radiopharmaceuticals (radium 223), selective estrogen receptor modulators (raloxifene), and immunosuppressants (methotrexate and corticosteroids).

When compared to antiresorptive medications, the level of evidence supporting other medication families as risk factors for MRONJ is level 5 (eg, isolated case reports or mini-case series [<5 cases]).¹⁴⁶⁻¹⁴⁹ Given that the poly-pharmaceutical management of cancer patients combined with the fact that cancer and immunosuppression are risk factors for MRONJ without exposure to antiresorptive agents, AAOMS believes that identifying a single medication as being the etiologic agent for MRONJ seems unlikely in case reports or mini-case series. Further controlled prospective studies will be required to measure the risk of MRONJ associated with non-antiresorptive agents.

MRONJ Risk Among Osteoporosis Patients

Most dentists and oral and maxillofacial surgeons evaluate patients in their practices exposed to antiresorptive therapy for management of osteoporosis (Table 1).

a. Risk for MRONJ among osteoporotic patients exposed to BPs.

The risk of MRONJ among study subjects assigned to placebo groups enrolled in osteoporosis clinical trials ranged from 0 percent to 0.02 percent.^{26,150,151} Among study subjects treated with BPs, the risk of MRONJ is 0.02 percent to 0.05 percent.^{37,75,152} Among patients exposed to IV zoledronate, the risk for MRONJ is estimated to be $\leq 0.02\%$ (≤ 2 per 10,000). For patients exposed to oral bisphosphonates, MRONJ risk is estimated to be $\leq 0.05\%$ (≤ 5 per 10,000).

b. MRONJ risk among osteoporotic patients exposed to RANK-L inhibitors.

After 10 years of follow-up, among patients exposed to DMB, the risk for MRONJ was reported to be 0.3 percent, almost an order of magnitude higher than for BPs.¹⁵³

c. The risk for MRONJ when exposed to romosozumab (0.03 percent to 0.05 percent) is comparable to alendronate (0.05 percent).^{37,151} In the placebo group, there were no cases of MRONJ.¹⁵¹ It will be important to continue to monitor romosozumab to assess its role as a risk factor for MRONJ.

The risk for MRONJ among osteoporosis patients treated with BPs ranges from 0.02 percent to 0.05 percent and overlaps the risk for MRONJ of patients enrolled in placebo groups (0 percent to 0.02 percent). The risk for MRONJ among patients treated with denosumab, however, has a larger range—from 0.04 percent to 0.3 percent. As such, additional research will be needed to better estimate the risk of MRONJ among patients receiving denosumab. The risk of MRONJ for patients exposed to romosozumab (0.03 percent to 0.05 percent) more closely aligns with the risk associated with BPs.^{37,151} However, given its recent introduction as a therapeutic agent, additional research will be needed to refine its association and risk estimate for MRONJ.

Based on this current review of data, the risk of developing MRONJ among osteoporotic patients exposed to BPs, DMB, and romosozumab is low. The occurrence of cases seen is best explained by a rare event among a large number of patients, 5.1 million over the age of 55, exposed to these drugs.¹⁵⁴

MRONJ Risk Among Patients with Nonmalignant Bone Disease

a. AAOMS identified two studies where DMB was used to manage aggressive giant cell tumors of bone.^{155,156} The risk of developing MRONJ in the two studies was broad and ranged from 0.7 percent to 5 percent. This is comparable to the risks of developing MRONJ in subjects treated with DMB for malignancies (range = 0 percent to 6.9 percent). Additional studies will be needed to confirm the risk estimate for MRONJ among patients with nonmalignant bone disease treated with antiresorptives.

b. There are very limited data describing the occurrence of MRONJ in the pediatric population for osteogenesis imperfecta and other conditions. In a systematic review estimating the risk of MRONJ among children with osteogenesis imperfecta, there were no cases of MRONJ identified in a sample of 486 subjects treated for 4.5 to 6.8 years.¹⁵⁷ In a different systematic review that estimated the risk for MRONJ among those under the age of 24 for several conditions treated using BPs, no cases of MRONJ were reported.¹⁵⁸

The overall quality of the studies included in both systematic reviews was limited by small sample sizes or lack of MRONJ-related risk factors.

Duration of Medication Therapy as a Risk Factor for MRONJ

Regardless of indications for therapy, the *duration of antiresorptive therapy* is a risk factor for developing MRONJ. Among cancer patients exposed to zoledronate or DMB (n = 5,723), the risk of developing MRONJ was, respectively, 0.5 percent and 0.8 percent at 1 year, 1.0 percent and 1.8 percent at 2 years, and 1.3 percent and 1.8 percent at 3 years.¹⁴¹ In a study by Saad et al, the investigators combined three-blinded phase three trials and found similar results, including a plateau after 2 years for patients exposed to DMB.⁵ In a more recent systematic review by Ng et al, the risk of MRONJ among cancer patients treated with zoledronate, was 1.6 percent to 4 percent after 2 years of treatment and 3.8 percent to 18 percent with more than 2 years of treatment.¹⁴⁵ Likewise, for DMB, the risks for developing MRONJ were 1.9 percent and 6.9 percent with <24 months and >24 months of exposure, respectively.¹⁴⁵

For patients receiving bisphosphonate therapy to manage osteoporosis, data regarding duration are mixed. Early on, the prevalence of MRONJ was reported as increasing over time from near 0 percent at baseline to 0.21 percent after four or more years of BP exposure based on retrospective analysis.^{152,159} More recent data from a large prospective, randomized placebo controlled trial demonstrate no significant increase in MRONJ in patients treated for up to 9 years.^{18,60,61} In addition, there are no postmarketing data or general clinical experience to support an MRONJ prevalence of 0.21 percent in any osteoporosis-treated group. Therefore, while duration may be a risk factor, the overall risk remains low.

LOCAL FACTORS

Dentoalveolar Operations

Dentoalveolar operations are the most common identifiable predisposing factor for developing MRONJ. Several studies report that among patients with MRONJ, tooth extraction is cited as a predisposing event ranging from 62 percent to 82 percent.^{5,75,160} While this information is important, it is not what most patients or clinicians want to know. Most providers and patients want an answer to the following clinical question: "Among patients exposed to antiresorptive medications, what is the risk for developing MRONJ following tooth extraction (or other dentoalveolar procedures such as implant placement or periodontal procedures)?" Cur-

rent estimates for the risk of MRONJ among osteoporotic patients exposed to BPs following tooth extraction range from 0 percent to 0.15 percent.^{161,162} For osteoporotic patients exposed to DMB, the risk for MRONJ following tooth extraction was 1 percent.¹⁶³

For cancer patients exposed to BPs, the risk of developing MRONJ after tooth extraction ranges from 1.6 percent to 14.8 percent.¹⁶⁴⁻¹⁶⁶ In a small case series, n = 61 subjects having 102 extractions, the risk for MRONJ after tooth extraction was 13.1 percent.¹⁶⁷ In a systematic review by Gaudin et al, the risk for MRONJ after tooth extraction (n = 564) was estimated to be 3.2 percent.¹⁶² While the estimates for developing MRONJ in high-risk patients undergoing tooth extraction vary, they cluster between 1 percent and 5 percent, similar to estimates of osteoradionecrosis following tooth extraction in irradiated patients.

The risk of developing MRONJ among patients who have been exposed to antiresorptive medications for other dentoalveolar operations such as dental implant placement and endodontic or periodontal procedures is unknown.¹⁶⁸ The risk for MRONJ after implant placement among patients treated with DMB has been reported to be 0.5 percent.¹⁶³ Absent better data, AAOMS cautions the use of these procedures in cancer patients exposed to antiresorptive therapies and recommends osteoporosis patients be informed of potential risks, albeit low, including development of MRONJ, early and late implant failure all of which have been described in case reports and clinical trials.

Anatomic Factors

Limited new information regarding anatomic risk factors for MRONJ is available. MRONJ is more likely to appear in the mandible (75 percent) than the maxilla (25 percent) but can appear in both jaws (4.5 percent).^{5,75} Denture use was associated with an increased risk for MRONJ among cancer patients exposed to zoledronate (OR = 4.9; 95 percent CI = 1.2 to 20.1).¹⁶⁹ In a study by Vahtsevanos et al, a sample of 1,621 cancer patients treated with intravenous zoledronate, ibandronate or pamidronate, there was a two-fold increased risk for MRONJ among denture wearers.¹⁷⁰

Concomitant Oral Disease

Pre-existing inflammatory dental disease such as periodontal disease or periapical pathology is cited as a risk factor.^{75,168} Among cancer patients with MRONJ, the pre-existing inflammatory dental disease was a risk factor among 50 percent of the cases.^{5,165}

Given that a common treatment of inflammatory dental disease is tooth extraction, pre-existing dental disease may confound the relationship between tooth

extraction and risk for MRONJ. Tooth extraction may expose MRONJ as opposed to being the precipitating event. It would be valuable to see an estimate of the association between tooth extraction and MRONJ adjusted for pre-existing inflammatory dental disease.

After tooth extraction and periodontal disease, the next most common risk factor is reported as “spontaneous” MRONJ with no identifiable dental risk factor.¹⁶⁸

DEMOGRAPHIC AND SYSTEMIC FACTORS AND OTHER MEDICATIONS

Age and sex are variably reported as risk factors for MRONJ.^{5,165,169-171} The higher prevalence of MRONJ in the female population is likely a reflection of the underlying disease for which the agents are being prescribed (eg, osteoporosis, breast cancer).

As noted previously, those under the age of 24 treated with antiresorptives for benign bone diseases have not demonstrated any risk for MRONJ even after an extended duration of therapy. The overall quality of the studies included even in systematic reviews is based on small sample sizes and the lack of other MRONJ-related risk factors. The risk of developing MRONJ in the pediatric population requires continued surveillance.

Corticosteroids are associated with an increased risk for MRONJ.^{5,168,171} There are concerns that corticosteroids increase the risk for MRONJ when given in conjunction with antiresorptive agents.

Comorbid conditions are inconsistently reported to be associated with an increased risk for MRONJ, including anemia (hemoglobin < 10 g/dL) and diabetes.^{5,171} Cancer type also is variably reported as a risk factor.^{170,172}

Tobacco use is variably reported as a risk factor for MRONJ. In a case-control study, tobacco use approached statistical significance as a risk factor for MRONJ in cancer patients (OR = 3.0; 95 percent CI = 0.8 to 10.4).¹⁶⁹ In a more recent case-controlled study, tobacco use was not associated with ONJ in a sample of cancer patients exposed to zoledronate.¹⁷¹ Vahtsevanos did not report an association between tobacco use and MRONJ.¹⁷⁰

In brief, after chemotherapy and corticosteroid exposure, the next most reported comorbidity is “no comorbidity.”¹⁶⁸

In summary, the current literature reaffirms that the risk of MRONJ is significantly greater in cancer patients receiving antiresorptive therapy compared to patients receiving antiresorptive therapy for osteoporosis. Moreover, the risk of MRONJ in osteoporosis patients receiving antiresorptive therapy continues to be very low regardless of drug type (BPs, DMB, romoszumab) or dosing schedule.

Management Strategies

TREATMENT GOALS

The major goals of treatment for patients at risk of developing or who have established MRONJ are:

- Prevention of MRONJ (see section [MRONJ risk among cancer patients](#) below).
- Prioritization and support of continued oncologic treatment in patients receiving antiresorptive therapy alone or in combination with immune modulators or antiangiogenic medications:
 - Oncology patients benefit from the therapeutic effect of antiresorptive therapy by controlling bone pain and reducing the incidence of other SREs.
- Prioritization and support of continued bone health and the prevention of fragility fractures
 - Patients with osteoporosis, osteopenia, and other metabolic bone diseases benefit from antiresorptive therapy by significantly reducing the risk of fragility fractures and other skeletal-related events.
- Preservation of quality of life through:
 - Patient education and reassurance.
 - Control of pain.
 - Control of secondary infection.
 - Prevention of extension of lesion and development of new areas of necrosis.

PREVENTION OF MRONJ

Numerous studies demonstrate potentially modifiable factors for reducing the risk of MRONJ, including performing high-risk surgical procedures prior to initiating therapy,^{95,173-175} using preoperative and postoperative antibiotics and antimicrobial mouth rinses,^{174,176-180} primarily closing extractions sites,¹⁷⁶⁻¹⁷⁸ and maintaining good oral hygiene.^{95,166,176,177,181} Maximizing overall patient health is always indicated, such as smoking cessation and diabetes optimization. Although no individual strategy nor collection of strategies eliminates all MRONJ risks, these preventive procedures are recommended.

The prevention of MRONJ begins with the realization that patients receiving antiresorptive therapies may have altered osseous wound-healing capacity, which may also be a risk for developing MRONJ. Similar to other common preventive strategies in medicine and dentistry, healthcare providers need to recognize the importance of coordinated dental care and pretreatment management in minimizing the risk of MRONJ. This requires a continuous effort to educate patients, dentists, and medical professionals

about the real risks associated with these therapies and clinical prevention paradigms that can mitigate MRONJ development.

AAOMS re-emphasizes the importance of a multidisciplinary approach to the treatment of patients who are receiving antiresorptive therapies. This may also apply to other immune modulators or targeted therapies taken alone or in combination with antiresorptives. This approach includes consultation with an appropriate dental professional when it is determined a patient would benefit from these therapies.

Optimization of Oral Health

The 2014 AAOMS position paper identified valid prophylactic treatment strategies that reduce the incidence of MRONJ. The efficacies of these strategies remain validated by subsequent studies that demonstrate the importance of pretreatment dental screening and regimented dental surveillance. There is a robust level of support for early screening and initiation of appropriate dental care prior to the initiation of antiresorptive therapy.^{38,182-186}

These preventive management strategies not only decrease the risk for MRONJ but accrue the benefits that all patients enjoy with optimum oral health.¹⁸⁶⁻¹⁹³

In a prospective study of prostate cancer patients with bone metastasis, instituting a more regimented dental health surveillance system resulted in a 2.5-fold reduction in relative risk compared to symptomatically driven dental treatment.¹⁸⁶ In a systematic review aimed at identifying prevention strategies associated with tooth extractions in patients at risk for MRONJ, no randomized clinical trials were reported.¹⁹⁴ However, there are many animal studies that demonstrate that periodontal or periapical inflammation plays a key role in creating a local environment that supports the development of bone necrosis in the context of systemic antiresorptive therapy.^{85,91,195,196}

Treatment planning for patients at risk of developing MRONJ should include a thorough examination of the oral cavity and a radiographic assessment when indicated. It is important to identify both acute infection and sites of potential infection to prevent future sequelae that could be exacerbated once drug therapies begin. Considerations during the clinical and radiographic assessment include patient motivation, patient education regarding dental care, fluoride application, chlorhexidine rinses, tooth mobility, periodontal disease, presence of root fragments, caries, periapical pathology, edentulism, and denture stability.¹⁹⁷

An additional benefit of early dental consultation, when the use of antiresorptive therapy is being considered, is that the patient is informed of the risk associated with these drug therapies and the risk incurred by not undergoing recommended

dental preventive measures before consenting to treatment.

Cessation of At-Risk Medication Therapy (Drug Holiday) Prior to Tooth Extraction or Other Procedures that Involve Osseous Injury (eg, Dental Implant Placement, Periodontal or Apical Endodontic Treatment)

The clinical practice of antiresorptive drug holidays to mitigate MRONJ risk in patients undergoing dentoalveolar surgery was controversial at the time of the previous AAOMS position paper in 2014 and remained the case in 2021. While the practice of a drug holiday has been accepted and recommended by several international professional societies,^{3,38,182,183,198} the evidence to support or refute such positions remains inconclusive. The difficulty in establishing or refuting the efficacy of drug holidays is due to the rarity of MRONJ in these patient populations. Therefore, since few events are reported, randomized-controlled trials provide insufficient data to create sound treatment protocols. In a 2020 systematic review that studied the efficacy of antiresorptive drug holiday in preventing MRONJ, a variety of papers were identified with differing conclusions suggesting that a high level of evidence for supporting or refuting the use of a holiday is missing.¹⁹⁹

The historical use of a drug holiday was intended to decrease the prevalence of MRONJ subsequent to the performance of high-risk surgical procedures. The concern regarding this practice is the loss of efficacy of antiresorptive therapy with the development of SREs and fragility fractures. Among others, factors for consideration may include disease-related risk (cancer vs osteoporosis), drug-dosing frequency, duration of therapy, comorbidities, other medications (especially chemotherapy, steroids, or antiangiogenics), degree of underlying infection/inflammation, and extent of surgery to be performed.

Of note, the working group was unable to reach a consensus regarding a recommendation on drug holidays and was evenly split between offering drug holidays to patients on a case-by-case basis using prior recommendations and those who never recommend drug holidays, believing that the risks of potential deleterious effects of suspending antiresorptive therapy may outweigh a benefit.

A special concern should be considered for suspending RANKL inhibitors in osteoporosis patients. Several studies have demonstrated a rebound increase in bone resorption following the discontinuation of DMB, resulting in an increased risk of multilevel vertebral fractures.²⁰⁰⁻²⁰² If DMB is to be suspended, the timing and duration of the holiday should be optimized in order to minimize this risk. The planned dentoalveolar surgery can be completed 3-4 months following the last dose of

Table 2. MRONJ PREVENTION STRATEGIES

Pretherapy (Nonmalignant Disease)	• Educate patient about the potential risks associated with long-term ART.* • Optimization of dental health can occur concurrent with ART.
Pretherapy (malignant disease)	• Educate patients about the higher risk of MRONJ and the importance of regimented dental care. • Optimization of the dental health <i>prior</i> to the initiation of ART if systemic conditions permit (extraction of nonrestorable teeth or teeth with a poor prognosis).
During antiresorptive therapy (nonmalignant disease)	• No alteration of operative plan for most patients. • Considerations include drug schedule, duration of therapy, comorbidities, other medications (especially chemotherapy, steroids, or antiangiogenics), degree of underlying infection/inflammation, and extent of surgery to be performed. Drug holidays are controversial. • BTM[†] are not a useful tool to assess MRONJ risk.
During antiresorptive therapy/targeted therapies (malignant disease)	• Educate patients about the higher MRONJ risk in the setting of malignant disease. • Educate the patient about the importance of regimented dental care and prevention. • Avoid dentoalveolar surgery if possible. • Consider root retention techniques to avoid extractions. • Dental implants are contraindicated. • Drug holidays are controversial.

* Antiresorptive therapies.

† Bone turnover markers (CTX).

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DMB when the level of osteoclast inhibition is waning. It can then be reinstituted 6-8 weeks postsurgery. This management strategy minimizes the length of the drug holiday while maintaining a favorable environment for bone healing.

Bone Turnover Markers. Since the 2014 AAOMS position paper, there has been a shift away from bone turnover markers. No biomarkers are validated for clinical decision-making, and continued research and prospective studies are required before these markers can be considered efficacious tools in estimating MRONJ risk.

Other Biomarkers

Biomarkers related to angiogenesis, VEGF activity, endocrine function, and PTH have more recently been described.²⁰³⁻²⁰⁵ These markers remain at an exploratory stage and are not yet validated for clinical decision-making.

PREVENTION STRATEGIES

Patients scheduled to initiate antiresorptive treatment for cancer therapy

The treatment objective for this group of patients is to minimize the risk of developing MRONJ (Table 2). Although a small percentage of patients receiving antiresorptives develop osteonecrosis of the jaw spontaneously, the majority of affected patients experience

this complication following dentoalveolar surgery.^{5,112,165,206,207} Therefore, *if systemic conditions permit*, initiation of antiresorptive therapy should be delayed until dental health is optimized.^{173,208} This decision must be made in conjunction with the treating physician and dentist and other specialists involved in the care of the patient. There is widespread consensus that optimizing dental health prior to initiating therapy is efficacious and of paramount importance.^{38,185,186,209} Medical oncologists should educate their patients about the importance of dental health and the efficacy of prophylactic dental treatment in the prevention of MRONJ. Similar to patients who are to receive radiation therapy, optimizing the dental health in patients receiving antiresorptives or other therapies that can compromise bone healing is essential. The pretreatment evaluation of dental health must extend beyond a review of systems and include a physical and radiographic exam. Therefore, a comprehensive dental examination performed by a dental health professional would be a prudent approach for all patients prior to receiving antiresorptive therapy for malignant disease. This level of dental health assessment is most appropriately performed by a dental health professional.

The importance of minimizing the burden of dental infection and inflammation prior to dentoalveolar surgery in this cohort of patients with an elevated MRONJ risk cannot be over-emphasized. Nonrestorable teeth

and those with a poor prognosis should be extracted. Other necessary elective dentoalveolar surgery also should be completed at this time. It remains advisable that antiresorptive therapy should be delayed, *if systemic conditions permit*, until the surgical site(s) have mucosalized or until there is adequate osseous healing. Dental prophylaxis, caries control, conservative restorative dentistry, and nonoperative endodontic therapy are critical to maintaining functionally sound teeth. This level of care must be continued on a frequent and indefinite basis.¹⁸⁵

The posterior lingual plate region is a common site for trauma and mucosal irritation in denture wearers.^{5,75,170} Therefore, patients with full or partial dentures should be examined for areas of mucosal trauma, especially along the lingual flange region. It also is critical that patients be educated as to the importance of dental hygiene and regular dental evaluations, and specifically instructed to report any pain, swelling, or exposed bone.

Patients scheduled to initiate antiresorptive treatment for osteoporosis

Patients who are scheduled to receive antiresorptive therapy for the prevention of fragility fractures assume a significantly lower risk of MRONJ. Therefore, the urgency and the timing of optimizing the dental health are not as crucial. However, at the initiation of treatment, it would be prudent to educate patients regarding the potential risks of MRONJ. The importance of optimizing dental health throughout this treatment period and beyond cannot be underestimated.

It is not uncommon for patients to seek the consultation of an oral and maxillofacial surgeon in guiding their decision about starting or continuing antiresorptive therapy. In this scenario, the consulting oral and maxillofacial surgeon should use this opportunity to place the risks and benefits into the proper perspective. More specifically, patients should be reminded of the benefits associated with antiresorptive therapies in preventing fragility fractures and an acknowledgment of the rare occurrence of MRONJ.

The initial enthusiasm and attention associated with the discovery of MRONJ have had unintended consequences. When initially described, a “class effect” was observed, suggesting that MRONJ rates for patients receiving oncologic doses of BPs and those receiving osteoporotic doses of BPs were similar. A plateau and a decline in the use of BPs for osteoporosis was noted in 2006 and is hypothesized to be associated with various safety concerns, such as MRONJ. Patients are becoming increasingly more reluctant to begin or comply with their antiresorptive therapy. Current evidence also confirms an increase in fragility fractures with significant associ-

ated morbidity. As one salient example, hip fracture rates in the United States declined each year from 2002 to 2012 and then plateaued at levels higher than projected for 2013 to 2015, attributable to an “osteoporosis treatment gap.”²¹⁰ Hip fracture carries significant morbidity, with only 40 percent to 60 percent of individuals recovering their prefracture level of mobility and ability to perform instrumental activities of daily living.²¹¹ These data are representative of a true health crisis. The documented risk for developing MRONJ is low; however, the patient-perceived risk is not. As such, patients are unwilling to start or continue antiresorptive medical therapy. Patients are irrationally denying themselves the tangible therapeutic benefit of antiresorptive therapy to minimize the risk of fragility fractures in order to prevent a minuscule risk of developing MRONJ.

It is clear the benefit of fracture prevention outweighs the risk of MRONJ development in osteoporotic patients.²¹² This benefit is even more favorable in the cancer population where bone-stabilizing medications significantly improve quality of life, and it is detrimental when antiresorptives are withheld due to MRONJ safety concerns.

Asymptomatic patients receiving antiresorptive therapies for cancer

Maintaining good oral hygiene and dental care is of paramount importance in preventing dental disease that may require eventual extractions or other dentoalveolar surgery. Procedures that involve direct osseous injury should be avoided if possible. If a dentoalveolar surgical procedure is unavoidable (eg, fractured tooth, advanced periodontal disease), patients should be informed of the associated risks. The benefit of a drug holiday remains unsubstantiated in this setting. Nonrestorable teeth may be treated by removal of the crowns and endodontic treatment of the remaining roots.²¹³ Teeth may be extracted if necessary. Placement of dental implants should be avoided in the oncology patient receiving parenteral antiresorptive therapy or antiangiogenic medications. Case series and systematic reviews have reported necrosis associated with antiresorptive therapy and implant placement.^{194,214-216}

Asymptomatic patients receiving antiresorptive therapy for osteoporosis

Since the 2014 position paper, epidemiologic data regarding the risk of MRONJ in patients receiving antiresorptive therapy for osteoporosis remain limited due to the lack of sound prospective studies with sufficient power. Nevertheless, the risk for developing MRONJ is between 0.02 percent and 0.04 percent for BPs and 0.3 percent for DMB. (see

Table 1). Sound recommendations based on strong clinical research design are still lacking for patients taking oral BPs.

In general, elective dentoalveolar surgery does not appear to be contraindicated in this group. Risk assessment for the development of MRONJ in these patients includes the above-stated data and the discussion above related to drug holidays.

The placement of dental implants in the context of antiresorptive therapy for osteoporosis continues to be an area of research interest. Several systematic reviews have acknowledged the lack of quality data and randomized clinical trials. Some studies have recommended caution, especially with a longer duration of therapy or steroid use.^{194,216} For example, in their systematic review, Granate et al,²¹⁶ identified several studies that reported an elevated MRONJ risk associated with implants placed in the posterior jaw if the duration of bisphosphonate therapy exceeded 3 years and if the patients were receiving systemic corticosteroids. In contrast to these studies, systematic reviews by Gelazius et al, and Stavropoulos et al, reported no increase in risk.^{214,217} A recent retrospective propensity-matched cohort study of 44,900 patients reported a decreased risk of ONJ in osteoporosis patients receiving implants compared to matched controls who did not have implants. Of note, 9,738 patients had a history of BP use, and the results for implants was in contrast to risk increase for patients who underwent tooth extraction.²¹⁸

Reports of implant-related (MRONJ) necrosis can be divided into the early (implant surgery-triggered) or late (implant presence-triggered) category.^{215,219,220} In these reviews, the majority of the implant-related necrosis were not related to the initial implant surgery but occurred late (>12 months) and often at sites where implants were placed prior to the initiation of bisphosphonate therapy. The common presentation was an en bloc failure, where the osseointegration of the implants is maintained within the sequestrum.^{220,221} This has been recognized as a separate pattern of failure that is distinct from the common peri-implantitis failure and considered by some to be pathognomonic of MRONJ. Although there are no prospective studies or systematic reviews pertaining to implant-related necrosis associated with RANKL inhibitors or other targeted therapies, AAOMS considers this to have a similar level of risk.

In summary, robust data do not exist, and available data are conflicting. Therefore, AAOMS suggests that if dental implants are placed, informed consent should be provided to include the low risk of MRONJ, as well as early and late implant failure. These patients should be placed on a regular long-term recall schedule.

Treatment Strategies

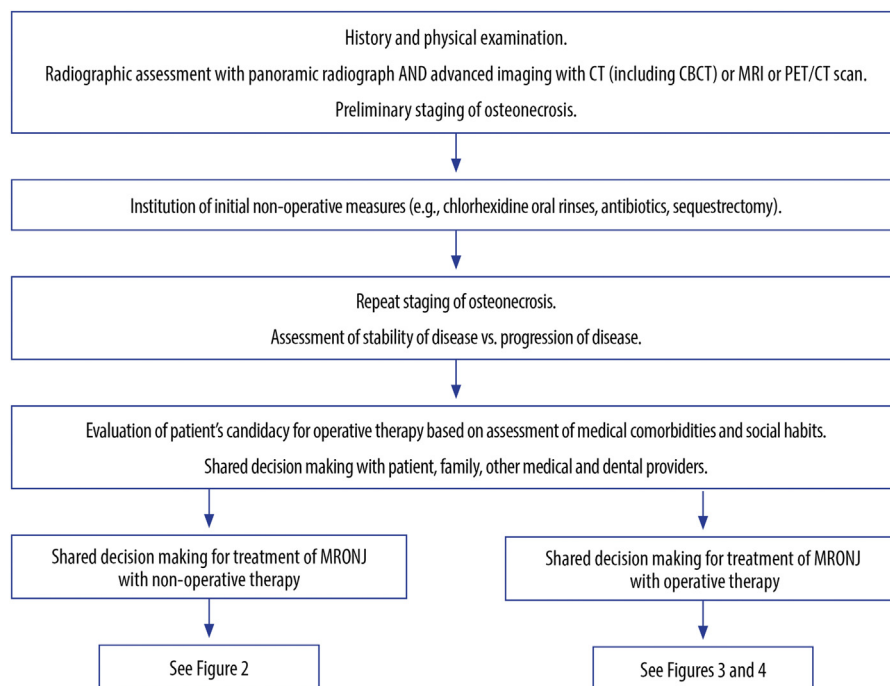
AAOMS has developed a series of treatment algorithms to streamline the evaluation (Fig 1) and management strategies (Figs 2-4) for patients with MRONJ. These strategies are based on a current review of nonoperative and operative therapies and their associated outcomes. Emphasis is placed on both nonoperative and operative management being acceptable for all stages of disease based on surgical judgment and patient factors in a shared decision-making model.

Nonoperative therapy

The efficacy of nonoperative therapies in the management of MRONJ is documented in the literature and provides a useful adjunct to the spectrum of management strategies that also include operative treatment (Fig 2). Nonoperative strategies can be useful in all stages, especially where significant comorbidities preclude operative treatment. They may also result in stabilization of disease or cure in earlier stages. The goal of both operative and nonoperative therapies remains the same: curative therapy and quality-of-life improvement. Nonoperative therapy heavily focuses on patient education, reassurance, control of pain, and control of secondary infection to allow for sequestration of the exposed, necrotic bone.^{3,63}

Decisions on operative versus nonoperative therapy should be patient-specific and tailored to individual needs. The risk versus benefit ratio (including quality of life with their current symptomology), ability to perform good wound care to prevent infection and disease spread, morbidity from a major surgical procedure, as well as oral function or dental rehabilitation after marginal or segmental resection should be considered. Radiographic imaging is of utmost importance in the evaluation of MRONJ lesions. Three-dimensional imaging can identify forming or fully formed sequestra and potentially decrease the invasiveness of a surgical procedure. Maintenance of maxillary or mandibular integrity is desirable, as the reconstruction of surgical defects in this population can be challenging.^{63,222}

Stage 1 patients can be managed with chlorhexidine wound care and improved oral hygiene to remove the biofilm from the necrotic bone surface.⁶³ Surgery may not be indicated in the absence of disease progression, with patient adequate quality of life.^{63,223} Stage 2 patients may struggle with local wound care and may require antibiotics for symptom control. Those patients who remain refractory to nonoperative treatment or those patients who cannot maintain adequate hygiene may benefit from operative therapy. In the presence of developing or established bony

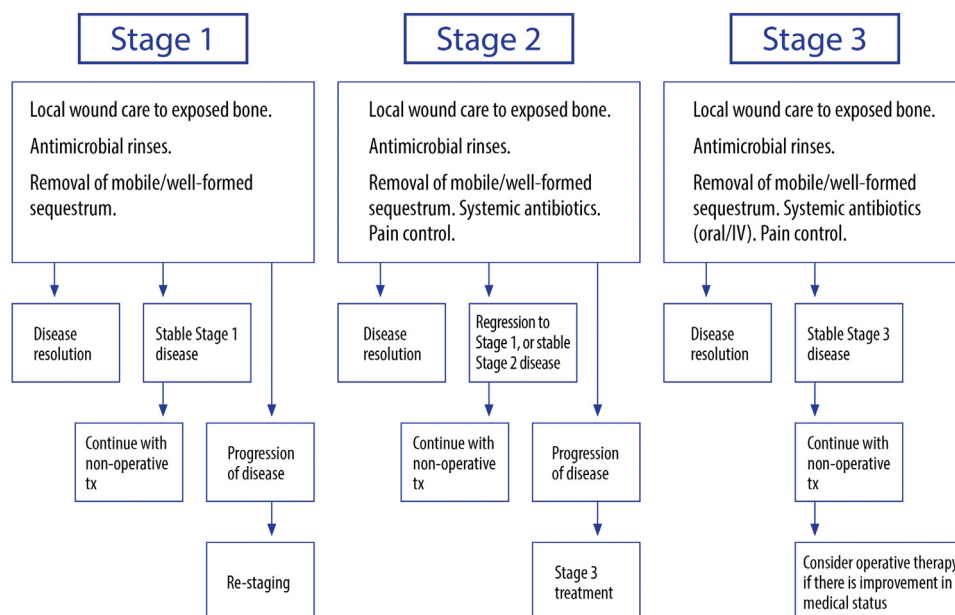
**FIGURE 1.** Initial evaluation.

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sequestra, nonoperative therapy may be indicated to allow for ultimate sequestrectomy. Exfoliation of the exposed, necrotic bone will often result in disease resolution.^{63,224,225} Therefore, for those patients with

Stage 2 or 3 diseases who are poor surgical candidates, nonoperative therapies may be indicated (Fig 2).

There is little evidence to suggest that the use of adjunctive therapies, such as hyperbaric oxygen or

**FIGURE 2.** Non-operative therapies.

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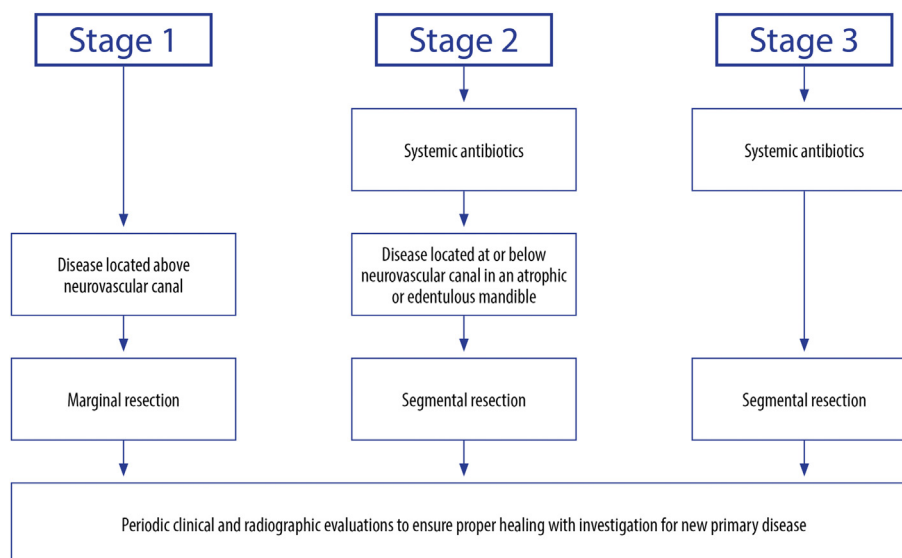


FIGURE 3. Operative therapies for mandibular disease.

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ozone therapy, can lead to MRONJ resolution. Larger studies and controlled trials have yet to demonstrate the efficacy of the aforementioned treatments.²²⁶⁻²²⁹ Therefore, these therapies should not be recommended as a mainstay of treatment at this time.

The use of vitamin E and pentoxifylline as an adjunct to standard MRONJ therapies have been reported only in case studies. A randomized, prospective, placebo-controlled trial of vitamin E and pentoxifylline is underway and will provide additional information about

this treatment modality. Teriparatide, one of the few anabolic agents used for the treatment of osteoporosis, also has shown promise as an adjunct for the treatment of MRONJ in osteoporotic patients.²³⁰

Operative therapy

While nonoperative therapy continues to be a treatment option for MRONJ, operative therapy is increasingly reported as a viable option with high success rates for all stages of the disease (Figs 3 and 4).

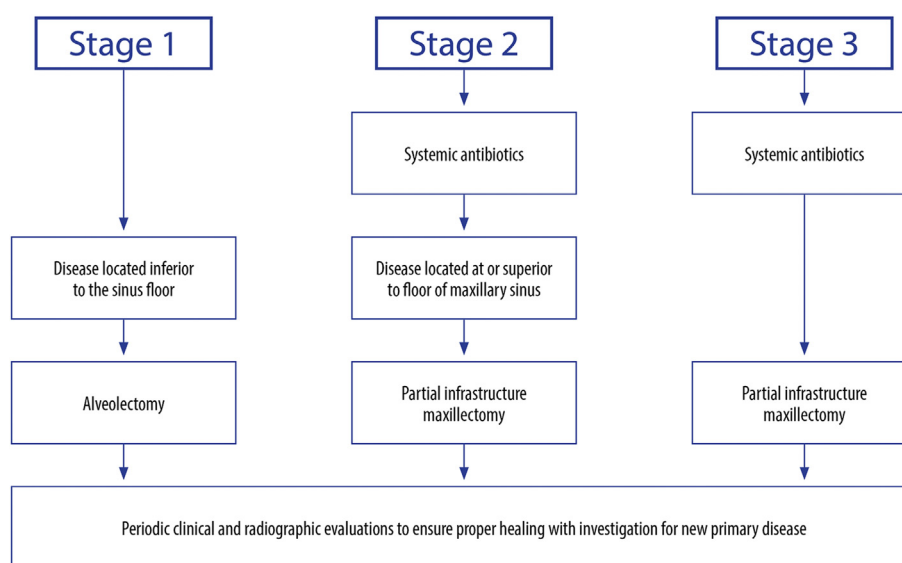


FIGURE 4. Operative therapies for maxillary disease.

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Numerous reports have identified high success rates associated with resection of MRONJ lesions.²³¹⁻²³⁷ Importantly, one must consider that MRONJ may progress over time, albeit in an unpredictable manner.²³⁸ Furthermore, adopting a nonoperative approach to MRONJ does not uniformly result in sequestration of the exposed necrotic bone with disease resolution.²³⁹ Thus, operative intervention should be explored and presented as a treatment option in an attempt to reduce the progression of disease with the recognition that early surgical intervention can predict beneficial patient outcomes.²⁴⁰

Segmental or marginal resection of the mandible and partial maxillectomy are effective methods to control MRONJ.^{231-238,241} This approach can be applied to patients with all stages of MRONJ, including Stage 1 disease.¹⁶⁹ These resections require margins beyond the borders of the necrotic bone to an area of vital, bleeding bone. Additional reports have identified success when surgical resection of MRONJ was performed by experienced surgeons.^{242,243} Consistent with surgical principles, control of comorbid conditions is paramount in managing MRONJ.²⁴¹ Physiologically compromised patients, such as those with an increasing burden of distant metastatic disease, may not respond favorably to resection of their osteonecrotic jaw, and may occasionally develop refractory disease.²⁴¹ Finally, surgical resection for MRONJ in patients with metastatic cancer may identify metastases in the jaw specimen, albeit in a minority of patients.⁶²

Active clinical and radiographic surveillance is critical in the nonoperative management of patients with Stage 1, 2, and 3 diseases to monitor for signs of disease progression. In patients who demonstrate the failure of nonoperative therapy, early operative intervention is recommended. In patients with a progressive clinical or radiographic disease or more advanced disease at presentation, surgical resection of MRONJ should be performed without first instituting prolonged nonoperative measures. MRONJ represents a complex wound whereby operative therapy can be performed in a timely fashion.^{241,244} Although controversy between operative and nonoperative therapies exist, operative treatment of patients has demonstrated maintenance of mucosal coverage, improved quality of life, and expedient resumption of antiresorptive therapy for all stages of MRONJ disease.²⁴⁵ The benefit of drug holidays for the operative intervention of MRONJ has not been substantiated.

Future Research

AAOMS realizes that MRONJ is a complex disease process with a multifactorial etiology for which

many questions remain unanswered. Continued pre-clinical and clinical data are required, especially in the form of prospective studies. Continued research efforts and the outcomes that result should be considered the foundation upon which recommendations are developed that will guide patients and providers. While the data supporting the conclusion that antiresorptives represent genuine risk factors are robust, this is not the case for other classifications of medications (eg, antiangiogenics, corticosteroids, immune modulators). Published studies have reported a relationship of certain dosing practices (eg, transition from BPs to DMB) or a synergistic effect between antiresorptive medications and antiangiogenic medication with a risk of MRONJ. These associations, however, are based on case reports and small case series. It also has been hypothesized that the total exposure to an antiresorptive medication is a risk factor for developing MRONJ. However, this has been difficult to demonstrate, possibly as a result of not having a good measure of exposure other than years of treatment. Similar to the cancer risk associated with tobacco use (eg, pack/years), the antiresorptive exposure risk MRONJ may be better defined as a cumulative dose load (eg, mg equivalent of BP/years of exposure) that would account for risk associated with different medications and dosing schedules over time. Dose-reduction protocols and individualized strategies for antiresorptive therapy in long-term cancer survivors with a metastatic bone disease are being explored. It remains to be determined if these protocols will reduce the risk of MRONJ in this patient cohort.²⁴⁶ AAOMS acknowledges the challenge of elucidating potential risks associated with non-antiresorptive therapies, alone or in combination with antiresorptive medications, and therefore considers it imperative that research efforts continue in the form of prospective studies.

A review of the current literature also failed to provide sound data in the form of randomized, controlled trials that would establish the effectiveness of biomarkers and drug holidays or validate a risk relationship with genetic markers and MRONJ. Until these relationships are established or refuted, AAOMS considers it prudent to recognize that these factors may play a role in the development and management of MRONJ.

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REVIEW ARTICLE

Italian position paper (SIPMO-SICMF) on medication-related osteonecrosis of the jaw (MRONJ)

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Abstract

Objective: This paper aims to describe the 2023 update position paper on MRONJ developed by the Italian Societies of Oral Pathology and Medicine (SIPMO) and of Maxillofacial Surgery (SICMF).

Methods: This is the second update following the 2013 and 2020 Italian position papers by the Expert panel, which is a representation of the two scientific societies (SIPMO and SICMF).

The paper is based on an extensive analysis of the available literature from January 2003 to February 2020, and the subsequent review of literature conducted between March 2020 and December 2022 to include all new relevant published papers to confirm or modify the previous set of recommendations.

Results: This position paper highlights the main issues of MRONJ on risk estimates, disease definition, diagnostic pathway, individual risk assessment, and the fundamental role of imaging in the diagnosis, classification, and management of MRONJ.

Conclusion: The Expert Panel confirmed the MRONJ definition, the diagnostic work-up, the clinical-radiological staging system and the prophylactic drug holiday, as recognized by SIPMO-SICMF; while, it presented novel indications regarding the categories at risk of MRONJ, the prevention strategies, and the treatment strategies associated with the therapeutic drug holiday.

KEYWORDS

bisphosphonate, bone density conservation agents, Denosumab, MRONJ, ONJ, osteonecrosis of the jaw

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1 | INTRODUCTION

Osteonecrosis of the jaw (ONJ) was initially reported only in association with bisphosphonate (BP) treatment (Marx, 2003). Clinical reports rapidly increased after the initial description of ONJ in metastatic bone cancer and multiple myeloma patients receiving BPs to include osteoporosis patients treated with oral BPs, inspiring the use of the term "Bisphosphonate related osteonecrosis of the Jaws" (BRONJ).

More recently patients receiving denosumab (DMB) and several biological agents (including but not limited to agents with antiangiogenic property) showed ONJ development (Otto et al., 2018). Starting from 2014, ONJ related to different medications has been grouped under the term Medication-Related ONJ (MRONJ) (Ruggiero et al., 2014).

The initial difficulties of clinicians who were involved in the diagnosis and treatment of a largely unknown disease are now being compensated by the growing body of knowledge on diagnosis, prevention and management (Bedogni, Fusco, et al., 2012; Colella et al., 2009; Fedele et al., 2010; Junquera & Gallego, 2008; Patel et al., 2012; Schiodt et al., 2014; Yarom et al., 2010).

Here, the authors present the 2023 position paper on MRONJ of the Italian Society of Oral Pathology & Medicine (SIPMO) and the Italian Society of Maxillo-Facial Surgery (SICMF).

The purpose of this position paper is to provide new insights on the following aspects of MRONJ: epidemiology, disease definition, diagnostic pathway (including the role of imaging), staging, risk assessment, preventive strategies and treatment algorithms. Also, this position paper has the aim to offer concise information for healthcare professionals who prescribe medications that increase the individual risk of MRONJ, and for oral healthcare providers.

2 | METHODOLOGY

The SIPMO-SICMF Expert Panel was established in 2010 under the auspices of the SIPMO and the SICMF to appraise the available literature and draft the Italian Recommendations on the diagnosis, prevention, and treatment of ONJ associated with BPs, which were released in Italian language in 2013 (Bedogni et al., 2013) and formally adopted by the Italian Ministry of Health the following year (Ministero Della Salute, 2014). The board panel comprised of a multidisciplinary group of clinicians and researchers with a special interest on ONJ. The Italian Recommendations were further revised to include all the relevant information on the new drugs associated with ONJ and the new categories of patients at increased risk and released in Italy in 2020 (Campisi, Bedogni, & Fusco, 2020). The 2020 update of the Italian Recommendations was endorsed by the following Scientific Societies: Italian Association of Medical Oncology

(AIOM), Italian Society for Osteoporosis, Mineral Metabolism and Bone Diseases (SIOMMMS), Italian Society of Medical and Interventional Radiology (SIRM), Italian Board of Medical Oncology Hospital Directors (CIPOMO), Board of University Professors of Oral Disciplines (CDUO), Italian Society of Osteoncology (ISO). The Recommendations were also endorsed by the National Dental Council Register of Italy and the Interuniversity National Consortium for Bio-Oncology.

This 2023 updated version of the Italian Recommendations was approved by all panellists and purposely written for publication in the English literature, to ensure broad dissemination.

For more details on the research methodology, refer to Supplements (Appendix S1).

2.1 | Definition

The up-to-date SIPMO-SICMF Expert Panel definition of Medication-Related Osteonecrosis of the Jaw (MRONJ), refers to MRONJ as "an adverse drug reaction described as the progressive destruction and death of bone that affects the mandible and maxilla of patients exposed to the treatment with medications known to increase the risk of disease, in the absence of a previous radiation treatment".

The initial attempt to define osteonecrosis of the jaws following exposure to BPs (BRONJ) was prompted by the American Association of Oral and Maxillofacial Surgery (AAOMS) in 2007 and 2009. The case definition of BRONJ was based on the presence of "exposed, necrotic bone in the maxillofacial region without resolution in 8–12 weeks in persons treated with a bisphosphonate who have not received radiation therapy to the jaws" (Advisory Task Force on Bisphosphonate-Related Osteonecrosis of the Jaws, 2007; Ruggiero et al., 2009).

American Association of Oral and Maxillofacial Surgery case definition of BRONJ was visibly incomplete and raised some criticisms as it was mainly based on the clinical evidence of exposed necrotic bone, leaving patients with signs of osteonecrosis other than bone exposure undiagnosed (i.e. non-exposed ONJ variant) (Bedogni, Fusco, et al., 2012; Colella et al., 2009; Fedele et al., 2010; Junquera & Gallego, 2008; Patel et al., 2012; Schiodt et al., 2014; Yarom et al., 2010).

AAOMS case definition was finally updated in 2014 to include "probing bone fistula" in the clinical presentation of the disease. The acronym also changed to MRONJ to embrace new antiresorptive medications and antiangiogenic drugs that have been linked to the development of ONJ. The AAOMS definition has not been changed since then (Ruggiero et al., 2014, 2022).

The SIPMO-SICMF Expert Panel initially recognized the limitations of the 2007 and 2009 AAOMS BRONJ definition and since



2012 proposed a comprehensive definition of BRONJ that was further updated in 2018 to include ONJ-related medications other than BPs (Bedogni et al., 2018; Bedogni, Fusco, et al., 2012).

Patients may be considered to have MRONJ if all the following characteristics are present: (Bedogni, Fusco, et al., 2012; Campisi, Mauceri, et al., 2020)

1. Current or previous treatment with bone-modifying agents (BMAs) and/or antiangiogenic agents (AAs).
2. Clinical and radiological findings of progressive bone destruction.
3. No history of radiation therapy to the jaws or the presence of primary oral malignancy or metastatic disease to the jaws.

Differential diagnosis should also disclose the presence of primary oral malignancy or metastatic disease (Arduino et al., 2015; Bedogni et al., 2007). Osteoradionecrosis should be suspected in patients with a history of radiation therapy to the jaws (Zadik et al., 2021).

The SIPMO-SICMF Expert Panel is concerned that the AAOMS case definition of MRONJ, which is mainly based on non-specific clinical aspects of the disease, may prove unreliable and suggests that MRONJ diagnosis should be reached through a differential diagnosis against different disorders with different aetiologies, presenting with similar clinical and radiological manifestations.

This position has been recently confirmed by the European Task Force on MRONJ, as patients may present with medical and oral conditions not to be confused with MRONJ (Schiodt et al., 2019).

2.2 | Epidemiology

Risk estimates of MRONJ are largely incomplete, biased and difficult to compare. Measures like incidence, prevalence, occurrence and frequency are often misinterpreted and generated data of limited value due to the inconsistency of available studies, short-term observation and the lack of cumulative long-term incidence.

Use of the 2007 AAOMS definition of MRONJ and to a minor extent that of 2014 to adjudicate confirmed MRONJ cases in large clinical trials on BMAs from the last 15 years could have contributed to underestimate the risk of developing osteonecrosis and keep the epidemiologic estimates low (Fedele et al., 2015; Fusco et al., 2011; Fusco, Mauceri, et al., 2022; Otto, Marx, et al., 2015; Schiodt et al., 2014).

Medication-Related Osteonecrosis of the Jaw risk profiles have gradually changed in the last two decades since the introduction of new associated medications to the market and the approval of supplementary indications for drugs already in use. Consequently, new categories of patients are being recognized at increased MRONJ risk.

At present, the SIPMO-SICMF Expert Panel recognizes four main categories of patients at increased MRONJ risk that are listed below:

1. *Cancer patients with Bone Metastases or Multiple Myeloma (BM/MM)*, commonly receiving monthly high doses of BMAs

(HD-BMAs) in combination or not with other drugs (e.g. chemotherapy, endocrine hormonal therapy, immunotherapy, AAs and other biological agents), or, rarely, receiving AAs alone. These patients present with several comorbid conditions, and show heterogeneous but relatively limited expected survival, although improved in recent years due to innovative therapies in some patient subpopulations (Fusco, Campisi, & Bedogni, 2022; Srivastava et al., 2021);

2. *Patients with osteoporosis (OP) and other non-malignant diseases* receiving low doses of BMAs (LD-BMAs). These patients are often elderly and with several comorbid conditions, but are likely to have longer expected survival than BM/MM patients (Kawahara et al., 2021);
3. *Cancer patients without bone metastases* receiving LD-BMAs to reduce the risk of non-metastatic bone fractures due to Cancer Treatment-Induced Bone Loss (CTIBL) in breast and prostate cancer patients, and/or to improve prognosis ("adjuvant" treatment of breast cancer patients) (Diana et al., 2021; Eisen et al., 2022);
4. *Patients with Giant Cell Tumour of Bone (GCTB)*; there are few but interesting data about GCTB patients treated for years with a monthly injection of DMB (HD-DMB), who display an increased risk of MRONJ (Palmerini et al., 2017).

In addition, several case reports and case series described in recent years the development of MRONJ in patients receiving AAs (anticancer drugs with antiangiogenic properties, including bevacizumab, aflibercept, inhibitors of tyrosine kinase and mammalian target of rapamycin inhibitors) or other biological drugs (agents for cancer or autoimmune diseases), with no history of BMA treatment. Yet, data are still inconsistent to quantify the risk of MRONJ for patients receiving only AAs (e.g. metastatic cancer patients receiving AAs alone without HD-BMAs; patients with rheumatic disorders receiving biological agents without LD-BMAs, etc.) (Fusco et al., 2016; King et al., 2019; Nicolatou-Galitis, Kouri, et al., 2019; Ruggiero et al., 2022; Suryani et al., 2022).

For more details, see Supplement, Appendix S2.

The SIPMO-SICMF Expert Panel agrees that BM/MM patients taking HD-BMAs are exposed to the highest risk of MRONJ, with incidence/prevalence ranging between 1% and >20%. Recent systematic and narrative reviews show a higher risk of MRONJ after zoledronic acid in comparison with other BPs (e.g. pamidronate, ibandronate) and a higher risk for DMB in comparison with zoledronic acid. MRONJ risk is thought to increase with time. In the case of BM/MM patients, Kaplan-Meier risk estimation curves show that MRONJ risk increases with the years of treatment (Fusco, Campisi, & Bedogni, 2022; Hata et al., 2022; Ikesue, Doi, et al., 2021; Ikesue, Mouri, et al., 2021; Loyson et al., 2018; Nakai et al., 2021; Rossi et al., 2021; Ueda et al., 2021). Whether the length of BMA treatment (i.e. duration and frequency of drug exposure) prevails over the observation time (i.e. patient survival) it is not fully understood. In any case, long treatment duration and prolonged survival rates raised the cumulative incidence up to 30% at 8-year observation in some metastatic cancer patient subgroups (Hata et al., 2022).

Clinical signs and symptoms

Abscess	Mucosal inflammation
Bone exposure	Non-healing post-extraction socket
Cutaneous fistula	Numbness of the lips ^a
Fluid discharge from the nose	Purulent discharge
Halitosis	Soft-tissue swelling
Intraoral fistula	Spontaneous loss of bone fragments
Jaw pain of bone origin	Sudden dental/implant mobility
Mandible fracture (fragment mobility)	Toothache
Mandibular deformation	Trismus

^aCaused by irritation of the inferior alveolar nerve and/or infraorbital nerve.

TABLE 1 Clinical signs and symptoms suspected for MRONJ (modified from Campisi, Bedogni et al, 2020).

On the contrary, the risk of MRONJ in OP patients receiving LD-BMAs is generally below 1%, with the exception of some subgroups at higher risk (e.g. patients with autoimmune or rheumatologic diseases). Real-life observation of the majority of MRONJ cases in OP patients after some years of LD-BMA treatment reflects the long-term results of the FREEDOM Extension study (<1%) (Watts et al., 2019).

The possible role of romosozumab in inducing MRONJ in OP patients is uncertain, as MRONJ was observed mostly in patients that also received bisphosphonates or DMB (see Supplement, Appendix S2).

Medication-Related Osteonecrosis of the Jaw incidence ranges between 0% and 5% in breast and prostate cancer patients treated with BMAs in the "CTIBL prevention" and "adjuvant" settings, but data are scarce and somewhat questionable (see Supplement, Appendix S2) (Brufsky et al., 2013; Coleman et al., 2013; Gnant et al., 2019; Smith et al., 2012).

Medication-Related Osteonecrosis of the Jaw incidence varies among GCBT patients receiving DMB, ranging between 1% and 13% (Jiang et al., 2022; Raimondi et al., 2020).

Additional studies are needed to confirm the risk estimates of MRONJ among CTIBL and GCBT patients.

Most relevant MRONJ quantitative estimates of patients at risk of MRONJ are detailed in Supplement (see Appendix S2).

The SIPMO-SICMF Expert Panel recognizes the existence of new populations of patients at risk of developing MRONJ, since the range of indications for the use of BMAs continues to expand and new agents are introduced in therapeutic armamentarium.

2.3 | MRONJ diagnosis

As previously mentioned, MRONJ has no unique definition, and controversies exist concerning the diagnostic process to be adopted (e.g. exposed vs. non-exposed variant, the role of imaging).

Ideally, both the definition and the diagnostic process of a given disease are intimately linked. Thus, a rigorous disease definition (i.e. case definition) can only lead to a definitive diagnosis when it is composed of signs and symptoms that are disease-specific (e.g. pattern recognition model). This is not the case with MRONJ or, which is known to present with several

non-specific clinical signs and symptoms, although some are more frequent than others (i.e. bone exposure and probing bone fistulas), as for other oral diseases like Sjogren's syndrome (Al Hamad et al., 2019).

When a case definition is unlikely to be exhaustive, which is the case of AAOMS MRONJ definition, a less restrictive definition should be adopted and a logical diagnostic workflow designed including clinical and instrumental examinations to identify all potential features of the disease and thus reducing the risk of misdiagnosis (e.g. hypothetical-deductive or analytical model) (Cawson et al., 2013; Rugarli, 2017).

Accordingly, the SIPMO-SICMF Expert Panel developed a diagnostic work-up of MRONJ that is based on the clinical and imaging features of the disease (Bedogni, Fusco, et al., 2012). The SIPMO-SICMF workflow is detailed in the next section.

2.3.1 | Clinical features of MRONJ

MRONJ is a multifaceted disease of the bone that can present with different clinical signs and symptoms (Table 1).

Despite the exposure of necrotic bone being the most common clinical feature of MRONJ, there are several other clinical signs and symptoms associated with MRONJ, including but not limited to probing bone fistula through the mucosa (Campisi, Bedogni, & Fusco, 2020; Ruggiero et al., 2014).

Among symptoms, the most frequently reported is pain, although it is absent at MRONJ onset in many patients (Bagan, Jimenez, et al., 2017; Ruggiero et al., 2022; Şahin et al., 2019; Schiodt et al., 2019). Numbness of the lips (e.g. paraesthesia of the lower lip, numb chin syndrome or Vincent's symptom) is also frequently reported by patients, but it is usually associated with advanced MRONJ (Miksad et al., 2011; Otto et al., 2009; Voss et al., 2018).

A peculiarity of MRONJ is that clinical signs and symptoms are not disease-specific, as they can be found in many other conditions, which makes the adoption of a purely clinical MRONJ case definition impractical in routine clinical care (Devlin et al., 2018).

Patients at risk for MRONJ can present with other common clinical conditions, including but not limited to plaque-related gingivitis/periodontitis, dental and periapical disease, benign fibro-osseous lesion of the jawbones, alveolar osteitis, chronic sclerosing



osteomyelitis and infectious osteomyelitis (Schiodt et al., 2019). Overall, these conditions need to be excluded to make a MRONJ diagnosis. Likewise, a differential diagnosis should also disclose primary oral malignancy and metastatic disease to the jaw (Arduino et al., 2015; Gander et al., 2014; Tocaciu et al., 2017).

Since MRONJ is a disease that mostly affects the jawbone architecture, imaging has long been considered a necessary part of the diagnostic process by the SIPMO-SICMF Expert Panel (Bedogni, Fusco, et al., 2012).

2.3.2 | Imaging features of MRONJ

Similar to clinical MRONJ features, radiological signs are not specific and cannot be used alone to diagnose MRONJ (Kim et al., 2020).

TABLE 2 Radiological signs suspected for MRONJ (modified from Campisi, Bedogni et al, 2020).

Radiological signs of MRONJ	Plain radiographs	CT-based radiological investigations
Cortical erosion	X	✓
Diffuse bone marrow sclerosis ^a	✓	✓
Focal bone marrow sclerosis ^a	X	✓
Opacified maxillary sinus	✓	✓
Osteolytic changes	X	✓
Osteolysis extending to the maxillary sinus	X	✓
Osteosclerosis of adjacent bones (zygoma and hard palate)	X	✓
Pathologic fracture	✓	✓
Periodontal space widening	✓	✓
Periosteal reaction	✓	✓
Persistent post-extraction socket	✓	✓
Sequester formation	✓	✓
Sinus tract ^b	X	✓
Thickening of the alveolar ridge	✓	✓
Thickening of the lamina dura	✓	✓
Thickening of the inferior alveolar nerve canal	✓	✓
Trabecular thickening	X	✓

Note: ✓, detectable. X, undetectable or detectable only in advanced MRONJ cases.

^aSclerosis: trabecular bone disorganization and poor corticomedullary differentiation.

^bOroantral, oronasal or orocutaneous communication due to bone destruction.

Several imaging features have been commonly associated with MRONJ, including but not limited to focal or diffuse bone marrow sclerosis, osteolytic changes, periodontal space widening, thickening of the inferior alveolar nerve canal, and sequester formation (Table 2).

Less clear is how to distinguish between early and late MRONJ signs. In fact, the ability to detect early bone changes in MRONJ is largely influenced by the imaging modality used (Wongratwanich et al., 2021).

No consensus has yet been reached regarding the imaging technique of choice to diagnose and screen MRONJ. Nevertheless, plain radiographs and computed tomography (CT) imaging have been adopted in many centres for their ability to image the underlying bone condition and display the radiological features of MRONJ in almost every patient (Baba et al., 2018; Bianchi et al., 2007; Bisdas et al., 2008; Hutchinson et al., 2010). Their wide availability, routine use among dental practitioners for the differential diagnosis of common dental conditions, ease of interpretation, limited contraindication and cost further support the use of plain radiograph and CT (Leite et al., 2014).

Dental x-rays and panoramic radiographs are the radiographic standard of care in routine dental practice and can help the dental practitioner to evaluate and definitively diagnose many oral diseases and conditions, with minimal radiation exposure (Leite et al., 2014; Malina-Altzinger et al., 2019; Ogura et al., 2021). Panoramic radiograph offers a fundamental understanding of the lesions and reveals bone changes suggestive of MRONJ (Cardoso et al., 2017; Moreno-Rabié et al., 2022), but it is much less accurate than 3D imaging (i.e. CT), especially in early disease onset (Guggenberger et al., 2014; Guo et al., 2016).

Magnetic Resonance Imaging (MRI) and other investigations, including bone scintigraphy or Single Photon Emission Computed Tomography (SPET-CT), may be useful adjuncts in selected complex cases (Bisdas et al., 2008; García-Ferrer et al., 2008; Huber et al., 2020; Watanabe et al., 2019).

The SIPMO-SICMF Expert Panel considers dental x-rays and panoramic radiographs useful first-line screening tools to differentiate among dental conditions that could mimic MRONJ in patients receiving BMAs and/or AAs. Nevertheless, the inclusion of second-line CT-based imaging modalities, such as Cone-Beam (CBCT) and multidetector computed tomography (MDCT), is essential to disclose early signs of MRONJ, anticipate diagnosis and correctly define stage and treatment options.

2.3.3 | Bone biopsy

Although the histology of bone can easily disclose necrotic bone from viable bone also in MRONJ patients, a bone biopsy is generally considered an unnecessary procedure (Campisi, Mauceri, et al., 2020). Location of the proper site for bone biopsy can be challenging in non-exposed MRONJ variants and it would require mucosal incisions to expose the underlying bone, amplifying the risk of bone exposure.

The use of bone biopsy in the diagnostic work-up of MRONJ may only be indicated to disclose the suspect of malignancy (Arduino

et al., 2015; Bedogni et al., 2007; Carlson et al., 2013; Mauceri et al., 2021; Terenzi et al., 2018; Tocaciu et al., 2017).

The SIPMO-SICMF Expert Panel favours a non-invasive diagnostic approach, in which bone biopsy should be provided only in the presence of suspected primary oral malignancy or metastatic bone disease in patients receiving medications at increased MRONJ risk.

2.3.4 | MRONJ case adjudication pathway

Overall, the SIPMO-SICMF Expert Panel recommends the combined use of clinical and radiologic signs to diagnose MRONJ, which likely increases the chance of a correct diagnosis by excluding common oral conditions that could be confused with MRONJ.

In this view, a “suspected MRONJ case” is defined by the presence of at least one clinical sign/symptom in a patient receiving BMAs and/or AAs. Instead, a “confirmed MRONJ case” is defined by the concomitant presence of at least one clinical sign/symptom suggestive of MRONJ and radiological signs of jawbone involvement in a patient receiving BMAs and/or AAs (see Table 2).

2.4 | Diagnostic workup

The SIPMO-SICMF Expert Panel endorses the adoption of a three-step diagnostic work-up of MRONJ as follows (Figure 1):

- **Step 1. Identification of a MRONJ suspected case.** Clinical suspicion should be raised whenever a patient receiving medications associated with MRONJ presents with signs and symptoms in the oral cavity like those described in MRONJ. Analysis of the patient's medical and dental history informs the existence of systemic and local risk factors that could be linked to the disease.
- **Step 2. Differential diagnosis of a MRONJ suspected case.** Oral examination can disclose several oral conditions, but supplemental radiological investigations are required to make the correct diagnosis. 2D imaging modalities (dental x-ray and panoramic radiograph) are useful tools to exclude the main dental and jaw bone conditions presenting clinical and radiological signs and/or symptoms overlapping with the initial phases of MRONJ.
- **Step 3. MRONJ case confirmation.** Every suspected MRONJ case that goes beyond the second step should be considered highly indicative of MRONJ. Second-line CT-based imaging (MDCT/CBCT) not only helps to confirm the diagnosis but it is also essential to accurately stage MRONJ and assign treatments accordingly.

Once the diagnosis of MRONJ is confirmed, reporting each case to the National Pharmacovigilance databases is recommended.

2.5 | Staging

Ideally, a sound staging system measures disease severity and identifies clusters of patients who require similar treatment and have similar

expected outcomes. Objective measures of disease severity and extension should be used to assess disease progression (Markson et al., 1991).

On purpose, the SIPMO-SICMF Expert Panel developed a three-stage clinical-radiological classification system of MRONJ centred on the presence of bone marrow sclerosis at CT-based imaging (Table 3) in adjunct to the patient's clinical findings (Bedogni, Fusco, et al., 2012).

Bone marrow sclerosis is the most frequent radiological sign of MRONJ, and it is also detected in the early phases of the disease (Kim et al., 2020). The SIPMO-SICMF staging system recognizes the presence of a focal disease stage (Stage 1), when bone marrow sclerosis is limited to the alveolar jawbone (Figure 2); a diffuse disease stage (Stage 2), when bone marrow sclerosis encompasses the basal bone (Figure 3); and a complicated stage (Stage 3), which comprises of diffuse bone marrow sclerosis along with clinical and radiological signs of advanced disease (Figure 4).

The presence of pain and purulent discharge does not translate into a worsened disease stage as they can manifest repeatedly through the course of the disease. For this reason, the SIPMO-SICMF staging system includes (a) asymptomatic and (b) symptomatic forms within the same disease stage. This prevents the so-called “ping-pong phenomenon”, which describes the cyclic transition (stage downgrading/upgrading) of MRONJ patients from one stage to another as a result of the antibiotics given to treat recurrent infection with associated pain (Bedogni, Fusco, et al., 2012).

Historically, AAOMS introduced in 2007 a classification system of BRONJ based exclusively on the clinical presentation of the disease (presence of exposed necrotic bone, pain, infection and clinical signs suggestive of complicated disease) to assign patients to different stages of disease severity (Stages 1–3) and treat it accordingly. While in current widespread use, the AAOMS staging system raised several criticisms since its initial publication, as many cases of MRONJ without exposed bone were being excluded from the diagnosis and treatment (Advisory Task Force on Bisphosphonate-Related Osteonecrosis of the Jaws, 2007).

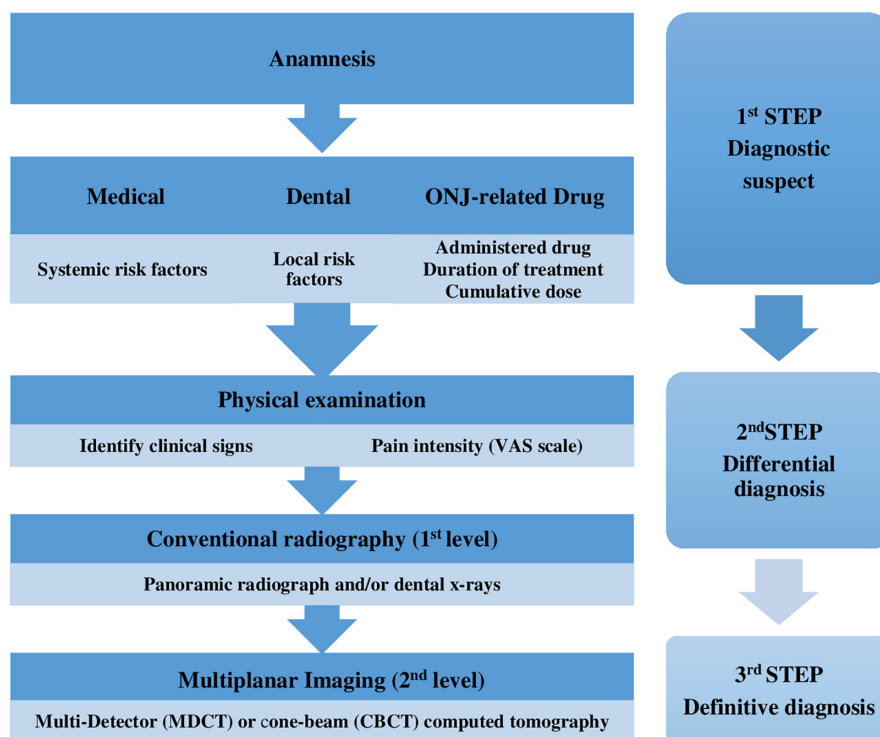
Despite two consecutive updates in 2009 (introduction of “Stage 0”) and 2014 (inclusion of patients with probing bone fistula in the running definition and classification system), AAOMS staging of MRONJ remained basically centred on the clinical presentation of the disease (Ruggiero et al., 2009, 2014).

The SIPMO-SICMF Expert Panel early recognized that the most significant weaknesses of the AAOMS staging system were the omission of radiological criteria to diagnose and classify the disease, and the underestimation of the real extent of bone involvement with the use of clinical signs and symptoms only (Bedogni, Fusco, et al., 2012).

Despite many authors having recommended the urgent need to include imaging criteria in the diagnosis and staging of MRONJ (Baba et al., 2018; Bedogni et al., 2014; Patel et al., 2012; Schiodt et al., 2014, 2019), the AAOMS Expert Panel maintained the current classification system with no apparent modifications in the latest 2022 update, where they formally rejected the idea of including radiological features, as these latter “may overestimate the true disease frequency by including false positives in the numerator” (Ruggiero et al., 2022).

The SIPMO-SICMF staging system has two major strengths when compared with the AAOMS classification: (i) it enables clustering of

FIGURE 1 Diagnostic Work-up of MRONJ (modified from Campisi, Bedogni et al, 2020).



patients with similar disease extent in the same stage and proper delivery of stage-related therapies with increasing intensity (Bettini et al., 2021); (ii) it considers clinical signs of recurrent bone infection and associated pain independent variables that cannot trigger any stage transition, as they do not correlate with the extent of bone involvement (Guo et al., 2016).

The SIPMO-SICMF staging system of MRONJ has remained unchanged since its introduction in 2012 and it has been increasingly used in Italy for the last 10 years (Albanese et al., 2020; Antonelli et al., 2021; Bacci et al., 2021; Bettini et al., 2021; Di Fede et al., 2016; Franchi et al., 2021; Marcianò, Ieni, et al., 2021; Marcianò, Ingrassiotta, et al., 2021; Mauceri et al., 2018; Migliario et al., 2017; Pignatelli et al., 2021).

The SIPMO-SICMF Expert Panel encourages the adoption of the proposed staging system on a large scale to assess MRONJ extent and to deliver stage-related treatments.

2.6 | Risk factors of MRONJ

Medication-Related Osteonecrosis of the Jaw is a multifactorial disease for which aetiology is not fully understood. While knowledge of risk factors associated with MRONJ continues to expand, the evolution of cancer therapies is likely to generate new patient populations at increased MRONJ risk. Several medical and dental comorbidities have been associated with an increased risk of MRONJ, while studies on more recent medications are still limited (King et al., 2019; McGowan et al., 2018; Nicolatou-Galitis, Kouri, et al., 2019; Suryani et al., 2022; Wick et al., 2022).

Medication-Related Osteonecrosis of the Jaw risk factors can be divided into three main groups: medication-related, systemic and local (Table 4) (Campisi et al., 2014; Ruggiero et al., 2014).

2.6.1 | Medication-related risk factors

There are two main classes of medications linked to MRONJ development:

1. Bone-modifying agents (BMAs): bisphosphonates (BPs) and denosumab (DMB);
2. Antiangiogenic agents (AAs): anti-VEGF (e.g. bevacizumab), TKIs (e.g. sunitinib) and mTORs (e.g. everolimus).

Other agents have been sporadically associated with MRONJ development but they still await confirmation from clinical studies (see Supplements, Appendix S2) (King et al., 2019; Suryani et al., 2022).

The following aspects are relevant to assess the medication-related risk: *agent type and dosing schedule, duration of treatment and cumulative dose*.

Bone-modifying agents therapy reduces osteoclastic activity, with a consequent decrease in bone resorption and inhibition of bone turnover. BMAs proved effective in preventing skeletal-related events in metastatic cancer patients and the risk of fragility fractures due to osteoporosis (Coleman et al., 2020; LeBoff et al., 2022).

Bisphosphonates (BPs), and nitrogen-containing BPs (NBPs) in particular, have a high affinity for bone and persist at the skeletal level for a long period of time (Fung et al., 2017).

MRONJ clinical-radiological staging system

Stage 1	FOCAL MRONJ: The presence of at least 1 clinical sign/symptom and increased bone density limited to the alveolar process at CT, w/ or w/o additional radiological signs. Clinical signs and symptoms: Abscess, bone exposure, halitosis, intraoral fistula, jaw pain of bone origin, mucosal inflammation, non-healing post-extraction socket, soft-tissue swelling, spontaneous loss of bone fragments, sudden dental/implant mobility, purulent discharge, toothache and trismus. CT signs: Trabecular thickening and/or focal bone marrow sclerosis, w/ or w/o cortical erosion, osteolytic changes, thickening of the alveolar ridge, thickening of the lamina dura, persistent post-extraction socket, periodontal space widening, thickening of the inferior alveolar nerve canal, sequester formation. Stage 1a: Asymptomatic (without pain) Stage 1b: Symptomatic (the presence of pain and/or purulent discharge)
Stage 2	DIFFUSE MRONJ: <i>The presence of at least 1 clinical sign/symptom and increased bone density extending to the basal bone at CT, w/ or w/o additional radiological signs.</i> Clinical signs and symptoms: same as Stage 1, plus mandibular deformation and numbness of the lips. CT signs: diffuse bone marrow sclerosis, w/ or w/o cortical erosion, osteolytic changes, thickening of the alveolar ridge, thickening of the lamina dura, persistent post-extraction socket, periodontal space widening, thickening of the inferior alveolar nerve canal, sequester formation, periosteal reaction and opacified maxillary sinus. Stage 2a: asymptomatic (without pain) Stage 2b: symptomatic (presence of pain and/or purulent discharge)
Stage 3	COMPLICATED MRONJ: <i>The presence of at least 1 clinical sign/symptom and increased bone density extended to the basal bone at CT, plus one or more of the following:</i> Clinical signs and symptoms: cutaneous fistula, mandible fracture, fluid discharge from the nose. CT signs: osteosclerosis of adjacent bones (zygoma and hard palate), pathologic fracture, osteolysis extending to the maxillary sinus, sinus tract (oroantral, oronasal fistula, orocutaneous). Stage 3a: asymptomatic (without pain) Stage 3b: symptomatic (presence of pain and/or purulent discharge)

TABLE 3 Clinical and radiological MRONJ staging system (modified from Campisi, Bedogni et al, 2020).

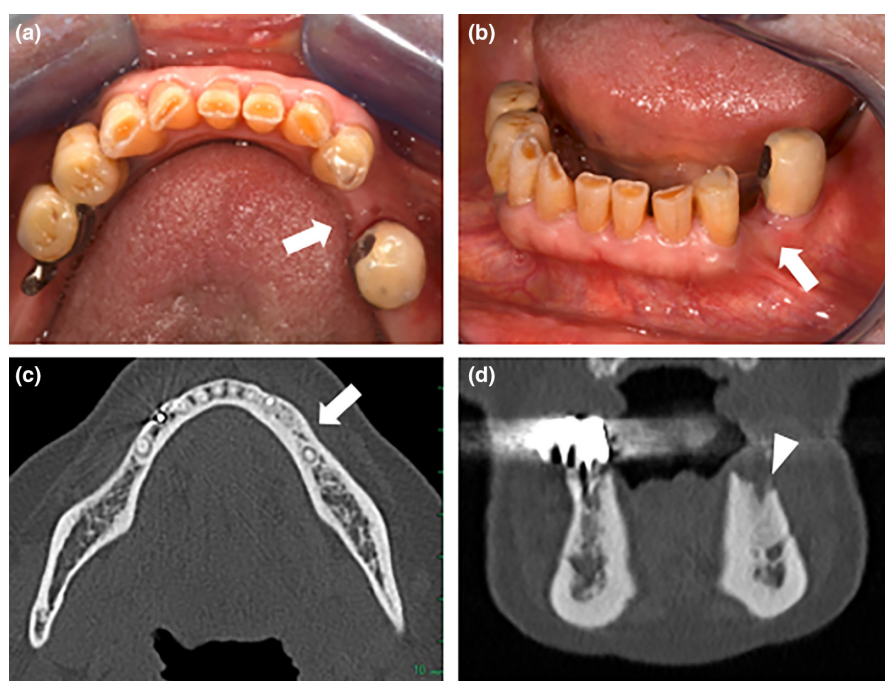


FIGURE 2 Stage 1, Focal MRONJ. Panel a: clinical view showing non-exposed MRONJ in the left posterior mandible (white arrow). Panel b: anterior view of the same lesion displays probing bone mucosal fistula (white arrow). Panel c and d: axial and coronal CT reconstructions show bone marrow sclerosis that is limited to the alveolar jawbone (white arrow and white arrow-head respectively).



FIGURE 3 Stage 2 (Diffuse MRONJ): Panel a: clinical view showing hyperaemic and swollen gingiva of the alveolar socket filled with granulation tissue (white arrow). Panel b: close-up view of the lesion (white arrow-head). Panel c and d: axial and coronal CT reconstructions show bone marrow sclerosis that reaches the basal bone; bone sequestration and periosteal reaction (white arrow and white arrow-head respectively).

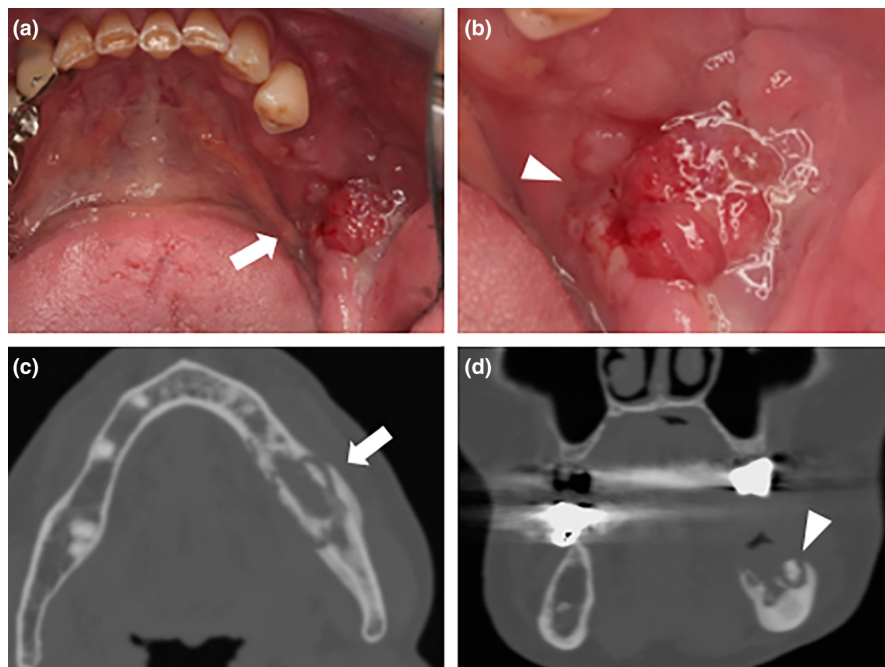
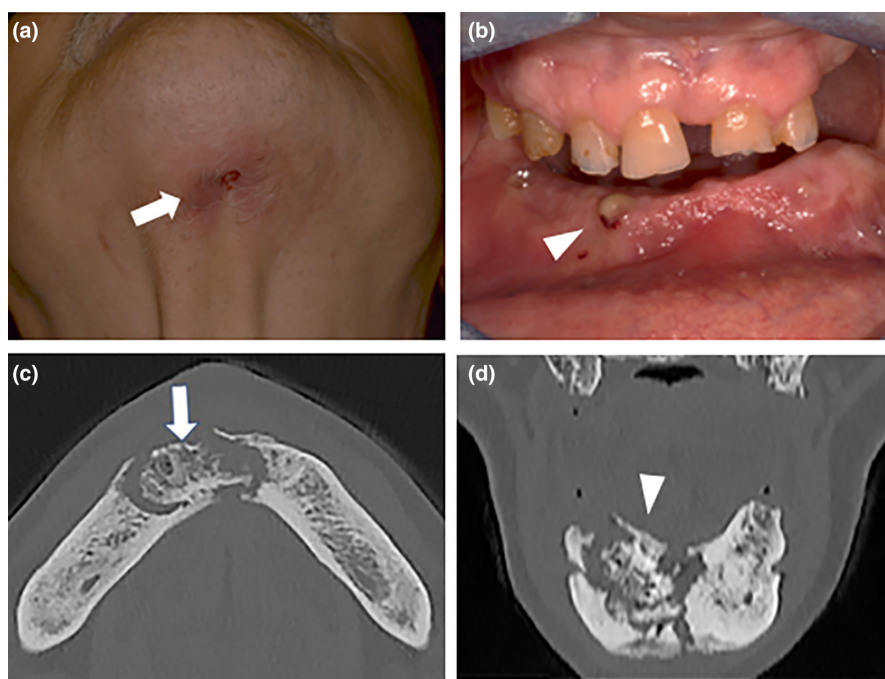


FIGURE 4 Stage 3, Complicated MRONJ. Panel a: clinical view showing a submental cutaneous fistula (white arrow). Panel b: intra-oral view showing single mucosal sinus track with purulent discharge at the level of the inferior gingiva (white arrow). Panel c and d: CT axial and coronal reconstructions display diffuse bone sclerosis encompassing the basal bone, sequester formation and pathologic mandibular fracture (white arrow and white arrow-head respectively).



In contrast, DMB does not accumulate in the bone and suppresses bone turnover through the inhibition of the receptor activator of the nuclear factor- κ B ligand complex (Fung et al., 2017; Mücke, Krestan, et al., 2016). DMB has a mean half-life of approximately 30 days and its effect on bone resorption gradually declines within 6 months after the last dose (Campisi et al., 2021; de Oliveira et al., 2016; Narayanan, 2013; O'Halloran et al., 2014; Otto et al., 2013).

AAs act against pro-angiogenic factors, with a modification of the mechanisms regulating bone repair and a reduction in bone remodelling (Saran et al., 2014). MRONJ development is rare in

patients treated with AAs while the risk of MRONJ likely increases when these drugs are coupled with BMAs (Srivastava et al., 2021). Also, AAs possess a short half-life (e.g. 20 days for bevacizumab, 40–60 hours for sunitinib) as compared with BMAs, and they seem to be characterized by a lower accumulative effect in the bone (Fusco et al., 2016; Maj et al., 2016).

The dosage and schedule of BMAs are major risk factors, with the oncological dosing of intravenous BPs (e.g. zoledronate 4 mg/month) and subcutaneous DMB (120 mg/month) being responsible for the highest risk of MRONJ as compared with low-dose oral/parenteral BPs and subcutaneous DMB (60 mg/6 month) given to

Drug-related	Agent type	• Bisphosphonates (BPs) • Denosumab (DMB) • Antiangiogenic drugs (AA)
	Dosage and schedule	
	Duration of treatment	
	Cumulative dose	
	Concomitant therapies	• Chemotherapy and anticancer hormone therapy • Corticosteroids • Immunotherapy • Medications inducing osteoporosis • Thalidomide
Systemic	Underlying disease	• Malignant disease • Osteoporosis
	Comorbidities	• Chronic kidney disease • Diabetes mellitus • Hypocalcaemia/Hyperparathyroidism and Osteomalacia/Vitamin D deficiency • Lifestyle habits (e.g. smoking) • Rheumatoid arthritis • Others
Local	Dental, periodontal, periapical and peri-implant infection	
	Dental implant surgery	
	Dentoalveolar surgery	
	Ill-fitting dentures (removable prostheses)	
	Anatomical variations	• Palatal and mandibular tori • Pronounced mylohyoid ridge
	Tooth extraction	

TABLE 4 Risk factors of MRONJ reported in the literature.

OP and non-metastatic cancer patients (Di Fede et al., 2016; Kuo et al., 2021; Liao et al., 2019; Watanabe, 2020).

The *duration of treatment* is also a relevant aspect to be considered. Most clinical studies on BM/MM patients receiving I.V.BPs report a median time to MRONJ development of 1.5–2 years, despite the great variability of the published data, and some cases of MRONJ occurred after many years of continuous treatment (Hata et al., 2022). In contrast, most cases of MRONJ in OP on BPs oral therapy usually occur after 2–3 years of treatment, with an average of 4.6 years (Fung et al., 2017).

While several studies report a similar time-to-onset of MRONJ for BM/MM patients receiving high-dose BPs and DMB, recent findings suggest that DMB-associated ONJ occurs earlier than BRONJ in cancer patients (Ikesue, Doi, et al., 2021; Loyson et al., 2018). In addition, the switch from BPs (i.e. zoledronic acid) to DMB significantly increases the risk of MRONJ development in BM/MM patients (see Supplement, Appendix S2) (Ikesue, Doi, et al., 2021; Loyson et al., 2018).

In BM/MM patients treated with monthly doses of BPs, the *cumulative dose* reflects the dosage and dosing intervals, the duration of treatment, and the affinity of any given agent for the target tissue. As the cumulative dose of BPs increases (above all with monthly administrations of zoledronic acid), the cumulative

incidence of MRONJ occurrence rises (Hata et al., 2022; Jadu et al., 2007).

At present, long-term estimates of MRONJ are scanty in BM/MM patients receiving monthly DMB injections (Fusco, Rossi, et al., 2022; Hata et al., 2022). Similarly, little data exist on the effect of cumulative dose of BMAs on the risk of MRONJ development in OP patients (Park et al., 2021).

Concomitant medical treatments, chemotherapy, anticancer hormone therapy and corticosteroids in particular, are reported as potential additional risk factors for MRONJ (Schiodt et al., 2018).

Since the introduction of a yearly I.V. infusion of Zoledronic acid and a six-monthly subcutaneous injection of DMB to prevent fragility fractures in OP patients, it has become clearer that the route of administration of BMAs plays a minor role in promoting MRONJ onset as compared with the other medication-related risk factors described. In fact, the frequency of MRONJ among OP patients remains much lower than for patients receiving high-dose BMAs (Everts-Graber et al., 2022; Fusco, Mauceri, et al., 2022).

Overall, the SIPMO-SICMF expert panel hypothesizes that the route of administration of BMAs may not to represent a major risk factor for MRONJ development, in the absence of strong scientific evidence.

Other known drug-related risk factors are described in Table 4.

2.6.2 | Systemic risk factors

Several malignant and systemic diseases have been closely associated with an increased MRONJ risk. BM/MM patients have the greatest susceptibility to MRONJ. Multiple myeloma and metastatic breast and prostate cancer are responsible for the large majority of MRONJ cases reported, followed by metastatic renal cell cancer. Many other cancer types were reported to be at increased risk of MRONJ development, which merits close attention in future clinical studies (King et al., 2019; McGowan et al., 2019). Also, GCTB has been recently associated with an increased risk of MRONJ (Jiang et al., 2022; Palmerini et al., 2017; Raimondi et al., 2020).

Osteoporosis is the most commonly reported non-malignant systemic disease associated with MRONJ onset; other conditions are rheumatoid arthritis, Sjogren syndrome and other autoimmune diseases (Di Fede et al., 2016; Fujieda et al., 2020; Kuo et al., 2021; Liao et al., 2019; Watanabe, 2020).

Systemic conditions for which BMAs and AAs are indicated can be a risk factor per se, although frequently associated comorbidities (e.g. diabetes mellitus, hypertension, etc) could contribute to the individual risk. This is particularly true for the elderly population and for cancer patients (King et al., 2019; McGowan et al., 2019).

Hypocalcaemia and hyperparathyroidism, osteomalacia and vitamin D deficiency have been linked to MRONJ development in the past years (Ardine et al., 2006; Bedogni, Saia, et al., 2012; Heim et al., 2017; Hokugo et al., 2010), but recent data oppose this hypothesis (Bedogni et al., 2019).

Other systemic risk factors have been reported to increase the risk of MRONJ, most of which still await validation (see Table 4).

The SIPMO-SICMF expert panel recommends clinicians record all medication-related and systemic risk factors of patients at increased MRONJ risk at first consultation, and keep their records updated for the entire period of surveillance.

2.6.3 | Local risk factors

Dental, periodontal and jaw bone conditions, which directly or indirectly compromise optimal oral health, increase the risk of MRONJ in a patient receiving BMAs and/or AAs (Table 4).

Tooth extraction has been traditionally considered the major local risk factor of MRONJ. The link between dental extraction and MRONJ development was referred to as the surgical trauma caused to the alveolar bone with a reduced bone turnover (Beth-Tasdogan et al., 2022; Gaudin et al., 2015; Khan et al., 2015; King et al., 2019; Kyrgidis et al., 2008; McGowan et al., 2019; Otto et al., 2012; Otto, Tröltzsch, et al., 2015; Schiodt et al., 2019).

Worthy of note, dental infections (apical/periodontal) are a common cause of dental extraction and have been reported to be associated per se with an increased risk of MRONJ development (Guarnieri et al., 2021; Lorenzo-Pouso et al., 2020; Tomasi et al., 2008).

Several experimental and clinical studies have shown the presence of early clinical, radiological and histological signs

suggestive of MRONJ in the alveolar socket of compromised teeth before or at the time the extraction takes place, suggesting that dental, periapical and periodontal infection may play an even more relevant role than the surgically induced bone trauma (Aghaloo et al., 2011; de França, 2017; Hadaya et al., 2019; Hasegawa et al., 2017; Jeong et al., 2017; Nicolatou-Galitis et al., 2015, 2020; Otto et al., 2021; Saia et al., 2010; Schiodt et al., 2019; Song et al., 2016).

Chronic dental, periapical and periodontal infections are generally associated with inflammatory responses that can alter osteoclast numbers and function through several pathways (e.g. direct stimulation of bone resorption, and stimulation of the release of inflammatory mediators). Overall, sites with increased bone turnover, such as extraction sites or areas of periodontal and periapical inflammation, are exposed to higher BMA intake which could explain the susceptibility of such areas to MRONJ onset (Aghaloo et al., 2011; Allen et al., 2011; Koerdt et al., 2014; Nicolatou-Galitis et al., 2015, 2020; Otto et al., 2021). Moreover, the presence of inflammatory cells in the alveolar bone seems to affect the physiological process of wound healing after the extraction of teeth with periapical or not treated periodontal diseases, in the presence of BMAs (Gkouveris et al., 2019; Landesberg et al., 2008; Soundia et al., 2018).

Similarly, it has been suggested that infection around dental implants may represent a notable risk factor for MRONJ development (Escobedo et al., 2020; Giovannacci et al., 2016; Kwon et al., 2014; Tempesta et al., 2021; Troeltzsch et al., 2016).

Although cases of implant surgery-triggered MRONJ have been reported soon after placement of dental fixtures in patients receiving HD-BMA therapy, the development of infectious-inflammatory processes around dental implants (e.g. peri-implantitis) is more likely to trigger MRONJ in patients who had been implanted well before the start of BMAs (Escobedo et al., 2020). The absence of a barrier effect at the bone-implant interface and the development of occlusal microcracks under masticatory load conditions may partly explain the occurrence of delayed MRONJ around dental implants (Kwon et al., 2014). That said, also OP patients receiving low-dose BPs are potentially exposed to the risk of delayed MRONJ development around dental implants after some years of therapy, but incidence/frequency is seemingly very low (Giovannacci et al., 2016; Kwon et al., 2014; Nisi et al., 2020).

Pressure sores from ill-fitting prostheses represent another established local risk factor for MRONJ.

Inadequate dentures that compress oral mucosa against bony prominences cause mucosal injury with subsequent bone exposure (Hasegawa et al., 2012; Levin et al., 2007).

Several cases of "spontaneous MRONJ" have been frequently described in the past to distinguish them from surgically triggered MRONJ (Khominsky & Lim, 2018; Vescovi et al., 2011). With the growing body of knowledge that most of those cases were triggered by dental and periodontal infections, reports of spontaneous MRONJ cases have become marginal. At present, a "spontaneous case" defines a patient

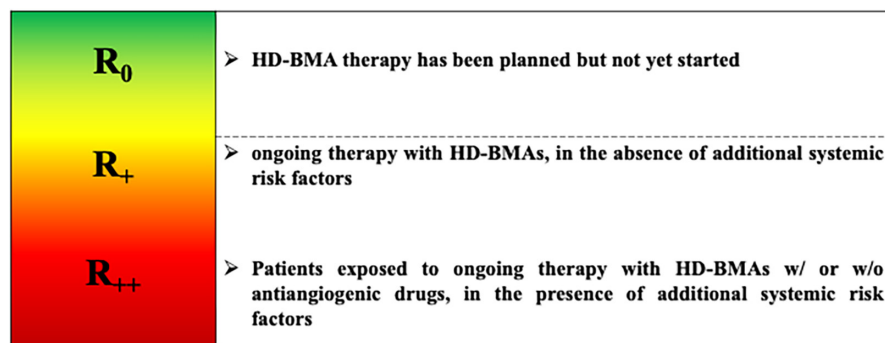


FIGURE 5 MRONJ risk gradient in patients receiving High-Dose (HD) BMAs (modified from Campisi, Bedogni et al, 2020). N.B. Local triggers are not used to define the risk gradient for each patient's category. Instead, they can precipitate MRONJ development when left unresolved. Patients at increased risk of skeletal-related events (SREs) who are shifted from HD-BPs to HD-DMB treatment represent a separate group where the cumulative dosage of the BP leads to the individual risk of MRONJ occurrence.

receiving medications associated with MRONJ, who presents with signs and symptoms of MRONJ in the absence of local risk factors. Genetic factors may play a role in these patients (Yang et al., 2019).

Other known local risk factors are described in Table 4.

In conclusion, the SIPMO-SICMF Expert Panel reaffirms that dental, periapical, periodontal and peri-implant infections are the main local risk factors for MRONJ, and it is concerned that many MRONJ patients in the past could have been misclassified as post-extraction cases, generating the idea that tooth extraction should be avoided in patients receiving BMA therapy.

Other groups of experts have come to the same (our) conclusion and support the idea that extraction of teeth with unfavourable prognosis has a clear role in MRONJ prevention, when properly and timely executed (Schiodt et al., 2019; Yarom et al., 2019).

2.7 | Individual risk assessment

Beyond the different levels of risk recognized for the outlined categories of patients receiving BMAs and/or AAs (see Section 2.2), individual risk assessment depends on the risk factors to which a single patient is exposed (Table 4).

Despite the lack of any reliable formula that can infer the individual risk of MRONJ occurrence at present, the SIPMO-SICMF Expert Panel believes that it is possible to grade the individual risk of MRONJ from time to time.

Although MRONJ cannot develop without exposure of patients to medications associated with MRONJ onset, medication-related risk factors, above detailed, are decisive to figuring out the individual risk assessment, in combination with systemic risk factors to which a single patient is exposed. Although local risk factors do not directly link to any risk category, they can trigger MRONJ development at any time if left untreated.

For the sake of better understanding, the SIPMO-SICMF Expert Panel agreed to showcase individual MRONJ risk separately for patients receiving HD-BMA and LD-BMA therapy.

2.7.1 | Patients at increased risk of MRONJ receiving high-dose BMAs

The SIPMO-SICMF Expert Panel considers that patients can be subgrouped based on the relative risk of MRONJ (R) (Figure 5) as follows:

1. HD-BMAs R_0 : antiresorptive therapy has been planned but not yet commenced.
2. HD-BMAs R_+ : ongoing antiresorptive therapy, in the absence of additional systemic risk factors. R_+ patients display an increased risk of MRONJ occurrence. MRONJ risk upraises in the presence of unresolved oral triggers.
3. HD-BMAs R_{++} : ongoing antiresorptive therapy, in the presence of additional systemic risk factors. MRONJ risk further upraises in the presence of unresolved oral triggers. These patients, including those also receiving antiangiogenic agents, display the highest MRONJ risk and should be carefully monitored. Based on the available literature, such risk is maintained a long time after drug discontinuation in the case of patients receiving BPs.

Within the limitation of the available evidence, the SIPMO-SICMF Expert Panel favours the inclusion of patients with Giant Cell Tumour of Bone (GCTB) on DMB therapy in the same at-risk category of patients receiving HD-BMAs, in light of the designated dosage and monthly schedule of BMA and the comparable frequency of MRONJ onset observed in these patients so far.

2.7.2 | Patients at increased risk of MRONJ receiving low-dose BMAs

Patients receiving low-dose BMA therapy can be stratified into two subgroups, based on the relative risk of MRONJ (R) (Figure 6):

1. LD-BMAs R_0 : patients who are about to start BMAs and patients who have been on BMA therapy for less than 3 years (current



or previous users), in the absence of additional systemic risk factors. R_0 patients display a MRONJ risk comparable to the general population (Campisi, Mauceri, et al., 2020; Di Fede et al., 2018).

- LD-BMAs R_x : patients who received BMAs for more than 3 years, and patients who received BMAs therapy for less than 3 years in the presence of additional systemic risk factors. R_x patients display an increased MRONJ risk that rises in the presence of unresolved oral triggers. This risk remains very low as compared with patients who are receiving HD-BMAs, although it cannot be quantified.

Worthy of note, non-metastatic breast and prostate cancer patients on hormonal therapy, who are being actively treated with low doses of BMAs for managing CTIBL or preventing cancer recurrence (adjuvant setting), should be included in the LD-BMA group risk stratification, in light of the small dosage and the prolonged dosing intervals (Diana et al., 2021).

2.8 | MRONJ prevention

To date, oral health preventive measures remain the most effective strategy to prevent MRONJ prior to, during and after the initiation of treatment with medications associated with an increased risk of MRONJ (Dimopoulos et al., 2009; Mücke, Deppe, et al., 2016; Owosho et al., 2018; Ripamonti et al., 2009).

The aim of primary prevention of MRONJ is to identify and remove all oral conditions that are known to trigger MRONJ and restore sound oral health. Primary prevention starts prior to and continues throughout the duration of therapy with ONJ-related medications; in the case of patients who receive BP therapy, oral surveillance and appropriate dental care should be prolonged after drug cessation, due to its known long-standing inhibition of jawbone remodelling (Campisi, Mauceri, et al., 2020; Schiodt et al., 2019). Unfortunately, the duration of oral surveillance after BP withdrawal is still unknown.

Additional targets of primary prevention are:

- Patient counselling, through which patients are informed about their individual risk of MRONJ and the indispensable adherence to timetabled oral check-ups. Counselling should also make patients aware of the possible clinical manifestations of MRONJ, thereby facilitating their timely recognition (i.e. early diagnosis) and rapid access to treatment;
- Enhanced communication between medical and oral health care providers to establish a beneficial interdisciplinary approach to at-risk patients (e.g. the timing of dental interventions, need for temporary drug withdrawal and drug restart).

When the goals of primary prevention are not targeted, poor adherence to antiresorptive treatments is likely since these medications are perceived as potentially dangerous by patients. This is particularly true for OP patients, whose poor adherence to osteoporosis treatment is a worldwide recognized concern, with a serious impact on deaths and hospitalization (e.g. repeated fractures) (Camacho et al., 2020; Inderjeeth et al., 2014).

One possible explanation for the poor OP patient's adherence to LD-BMA treatment is their fear of potential long-term side effects, including MRONJ occurrence. Awareness of medication-related risks and—most of all—knowledge of the risk reduction strategies proved effective to minimize the risk of MRONJ and might improve OP patients' adherence to antiresorptive treatment.

2.9 | Prevention strategies

MRONJ risk-reduction strategies have been developed in the past years and include the following dental treatments, which should be applied routinely to maximize the oral health of at-risk patients:

- Non-invasive dental procedures to maintain or restore oral health;
- Invasive surgical procedures, such as dental extraction of non-restorable and compromised teeth, to eliminate potential triggering factors.

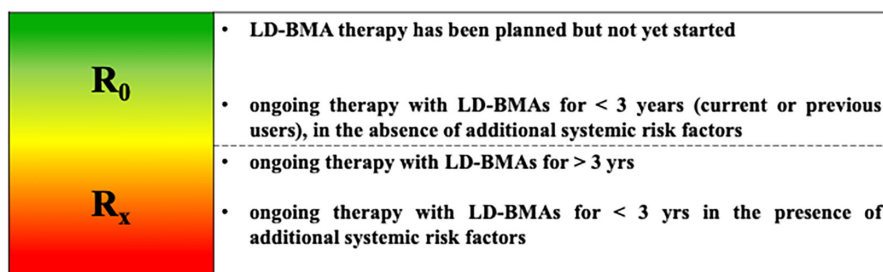


FIGURE 6 MRONJ risk gradient in patients receiving Low-Dose (LD) BMAs (modified from Campisi, Bedogni et al, 2020). N.B. Local triggers are not used to define the risk gradient for each patient's category. Instead, they can precipitate MRONJ development when left unresolved. Patients at increased risk of fragility fractures who are shifted from LD-BPs to LD-DMB treatment represent a separate group where the cumulative dosage of the BP leads to the individual risk of MRONJ occurrence.

Furthermore, it is considered crucial to practice a proper personal oral hygiene in order to reduce gingival inflammation and the possible development of periodontitis.

MRONJ risk-reduction strategies depend on the individual risks of patients, change over time and vary between different patient populations.

a. *Patients scheduled to start treatment with ONJ-related medications (HD-BMAs R_0 and LD-BMAs R_0 categories).*

Pre-treatment evaluation must be performed by an oral health care provider; it should include a comprehensive clinical and radiographic dental examination, the assessment of dental, periodontal and peri-implant status, quality of restorations and the inspection of dentures looking for areas of mucosal trauma. Timely management of potential triggers should be accomplished and initiation of antiresorptive therapy delayed until dental and periodontal health is optimized if systemic conditions permit. In the case of dental, periapical, periodontal or peri-implant infections that require invasive treatment at the bone interface, the patient should be strictly monitored and BMA treatment postponed until soft-tissue healing is achieved (Mauceri et al., 2022).

In the absence of conclusive data, the SIPMO-SICMF Expert Panel agrees with other International Recommendations on the importance of pre-treatment oral evaluation and supports the idea that patients about to start HD-BMA therapy with and without AAs or LD-BMA therapy require a separate approach, based on their distinctive MRONJ risk.

Patients about to start HD-BMAs with or without AAs should always undergo thorough dental and periodontal examination and management of potential infectious foci before initiation of BMAs, due to the growing risk of MRONJ onset. Hypercalcemia of malignancy represents an exception that requires immediate start of BMA treatment. In these patients, initial oral evaluation should be performed as soon as the hypercalcaemic state has been resolved.

Alternatively, a pre-treatment oral evaluation is not mandatory for patients about to start LD-BMA therapy, because they display low to null MRONJ risk as compared with the general population in the first few years of treatment (Campisi, Mauceri, et al., 2020).

The SIPMO-SICMF Expert Panel recommends pre-treatment oral evaluation for patients about to start HD-BMAs. Also, it considers counselling of patients who are scheduled to receive LD-BMAs at least as important as restoring and maintaining their oral health and recommends activating prevention measures within six months from the start of BMA therapy.

b. *Patients on active treatment with ONJ-related medications (HD-BMAs R_{++} , R_{++} and LD-BMAs R_x categories).*

Although periodic oral surveillance is accomplished by oral healthcare providers, it is also the responsibility of the treating physicians to contribute to the patient's adherence to the scheduled

recall visits, to minimize the risk of MRONJ occurrence and ensure the patient's persistence on antiresorptive medications.

Recall visits should be scheduled to maintain the oral health of patients for the entire duration of treatment with BMA, with or without AAs, and also periodically later after the end of the treatment. Patients who had received BP treatment should be maintained on strict oral health surveillance for a long time after drug discontinuation. Timely management of potential triggers should be guaranteed to patients (Campisi, Mauceri, et al., 2020; Mauceri et al., 2022).

Furthermore, the prevention measures should be planned also in patients after the end of the treatment with ONJ-related medications and in patients that already suffered from MRONJ.

The SIPMO-SICMF Expert Panel recommends maintaining oral surveillance for the entire duration of DMB treatment and much longer in case of BP therapy, with periodic recall visits on a 4-month basis for HD-BMAs R_{++} , R_{++} categories and every 6 months for LD-BMAs R_x category.

2.9.1 | Dental management

Dental treatment includes *essential or emergent procedures* aimed at removing infectious triggers (e.g. pulpitis, pericoronitis, osteitis, dental or periodontal abscess, peri-implantitis, dental trauma, extensive caries or defective restorations that cause pain or tissue damage, adjustments in dentures that cause damage to oral structures) and *non-essential or elective procedures*, which include but are not limited to cosmetic procedures, orthodontic therapy, replacement of amalgam restorations for aesthetic reasons, elective periodontal care, intentional root canal treatment, prosthodontics and elective oral surgery. A major proportion of the dental treatments provided to the general population are elective in nature.

The SIPMO-SICMF Expert Panel classifies dental treatments into the following categories based on a risk/benefit ratio for patients; a traffic light colour code is used for the sake of simplicity (Table 5):

1. *Indicated treatments* (green light): all essential procedures required to treat emergent oral conditions and those elective non-surgical procedures that have not been associated with an increased MRONJ risk;
2. *Feasible treatments* (yellow light): those elective procedures with uncertain MRONJ risk under specific conditions;
3. *Contraindicated treatments* (red light): those elective surgical procedures that are linked to a clearly unfavourable risk/benefit ratio.

In brief, all non-surgical procedures that are essential for the resolution of infectious processes (e.g. restorative dentistry, endodontics and periodontal therapy) are clearly indicated in all patients receiving BMA therapy with or without AAs, independent from their individual MRONJ risk and they should be delivered as soon as possible. The successful restoration/preservation of salvageable teeth is

likely to reduce the need for surgical therapies and the risk of overt MRONJ (Di Fede et al., 2018).

Teeth with poor prognosis or that have failed to resolve with restorative treatment should not be declined dental extraction (Nicolatou-Galitis et al., 2020; Otto, Tröltzsch, et al., 2015; Saia et al., 2010; Schiodt et al., 2019).

Overall, since chronic infection is the main local risk factor for MRONJ, tooth extraction has a clear preventative role of MRONJ, when properly and timely executed (Schiodt et al., 2019).

While HD-BMAs R_0 and LD-BMAs R_0 patients may be safely subjected to routine dental extraction, HD-BMAs R_+/R_{++} and LD-BMAs R_x patients should undergo dental extraction using specific surgical protocols that include mucoperiosteal flap elevation, atraumatic tooth extraction, alveolectomy and smoothing of bone edges and tension-free soft tissue closure (Campisi, Mauceri, et al., 2020; Di Fede et al., 2018; Heufelder et al., 2014; Ristow et al., 2021). Biopsy of the alveolar bone to assess bone viability may be considered in patients at increased MRONJ risk at the time of dental extraction (Nicolatou-Galitis, Schiodt, et al., 2019; Saia et al., 2010). In the case of dental extraction, perioperative administration of systemic antibiotics is often prescribed to lower MRONJ risk in HD-BMAs R_+/R_{++} and LD-BMAs R_x patients. A recent systematic review aimed at assessing the effectiveness of antibiotic therapy in preventing MRONJ in patients receiving BMA and/or AA therapies, when in need of teeth extraction. The authors found that perioperative antibiotic regimens reported in the literature are mainly empirical and lack validation. Peroral penicillin-based antibiotic therapy, either alone or accompanied by a β -lactamase inhibitor or metronidazole was found the most used perioperative protocol (Cabras et al., 2021).

The SIPMO-SICMF Expert Panel recommends the timely extraction of non-restorable teeth and the adoption of specific risk reduction strategies depending on the individual MRONJ risk profile.

Dental implant placement is generally contraindicated in patients scheduled for or already receiving HD-BMA therapy (Campisi, Bedogni, & Fusco, 2020; Japanese Allied Committee On Osteonecrosis Of The Jaw et al., 2017; Yarom et al., 2019).

Alternatively, implant placement is feasible in patients scheduled for or already receiving LD-BMA therapy, irrespective of the agent type and route of administration. Yet, these patients should be informed about the low albeit non-quantifiable risk of MRONJ onset (Escobedo et al., 2020; Giovannacci et al., 2016; Kwon et al., 2014).

Implant-related MRONJ has been recently classified into early (implant surgery-triggered) or late (implant presence-triggered), with the latter occurring most frequently and at sites where implants were placed prior to the initiation of BMAs (Giovannacci et al., 2016; Stavropoulos et al., 2018).

The SIPMO-SICMF Expert Panel recommends avoiding implant placement in patients scheduled for or already receiving HD-BMA therapy, while it considers dental implant surgery a feasible procedure in patients undergoing LD-BMA therapy. In any case, the SIPMO-SICMF Expert Panel recommends evaluating each case individually, performing a complete risk assessment and warning patients scheduled for or already receiving LD-BMA therapy on the potential threat of late MRONJ onset as well as describing all alternative strategies for the restoration of oral functions. Informed consent should be obtained on the possible long-term risks and benefits of the procedure.

2.9.2 | Diagnostic laboratory tests

Literature has provided insufficient data to support the use of biomarkers to predict MRONJ risk in patients on antiresorptive medication who need oral surgical procedures (Dal Prá et al., 2017; Don-Wauchope & Cole, 2009; Peisker et al., 2017; Traboulsi-Garet

TABLE 5 Dental management of patients who receive BMAs and are at increased MRONJ risk (modified from Campisi, Bedogni et al, 2020).

	Dental treatments	HD-BMAs R_+ and R_{++} patients	LD-BMAs R_x patients
Non-surgical procedures	Restorative dentistry	Indicated	Indicated
	Endodontic treatment	Indicated	Indicated
	Orthodontic treatment	Feasible	Feasible
	Periodontal therapy	Indicated	Indicated
	Prosthetic rehabilitation	Feasible	Feasible
Surgical procedures ^a	Dentoalveolar surgery	Indicated	Indicated
	Tooth extraction	Indicated	Indicated
	Pre-implant bone surgery	Contraindicated	Feasible
	Dental implant surgery	Contraindicated	Feasible ^b
	Periodontal surgery	Indicated	Indicated
	Endodontic surgery	Indicated	Indicated

^aTight soft-tissue closure must be ensured. Except for LD-DMB R_x patients who do not necessitate drug suspension before surgery, BMAs should be resumed once wound healing has been achieved (4–6 weeks).

^bIt is advisable to inform the patient about the long-term risk of implant-triggered MRONJ.

Green shade are indicated treatments. Yellow shade are feasible treatments; while in red shade there are contraindicated treatments.

Molecule type	Last dose before surgery	Resume treatment
Medication withdrawal in HD-BMAs R_+ and R_{++} patients		
Bisphosphonates	1 week	Once wound healing has been achieved (4–6 weeks after surgery)
Denosumab (Xgeva®)	3 weeks	
Bevacizumab	5–8 weeks	
Sunitinib	1 week	
Everolimus	1 week	
Medication withdrawal in LD-BMAs R_x patients		
Bisphosphonates	1 week	Once wound healing has been achieved (4–6 weeks after surgery)
Denosumab (Prolia®)	No need for suspension ^a	

^aElective surgical treatments including tooth extraction can be preferably performed without restrictions 5 months after the last dose of Prolia®, taking advantage of the reactivated bone turnover and of the recovered healing capacity of bone. This “window of opportunity” lasts about 2 months. A planned 1-month delay of the scheduled dose of DMB may be enough to foster soft-tissue healing, without added risk of fragility fractures. Instead, non-deferrable extraction of compromised teeth should be performed from the 3rd week after the last dose of Prolia® adopting specific risk reduction protocols, which include tight soft-tissue closure.

et al., 2022). Currently, no biomarker with MRONJ specificity and sensitivity exists. Biomarkers should be interpreted in relation to the patients' clinical, radiological and systemic conditions (Moraschini et al., 2019).

The SIPMO-SICMF Expert Panel agrees that the use of biomarkers to predict MRONJ onset could be misleading and does not favour their clinical use until proven otherwise.

2.10 | Prophylactic drug holiday

Planned interruption of BPs, in the absence of MRONJ, before oral surgical procedures (prophylactic drug holiday) including tooth extraction has been emphasized in the past 20 years to prevent impaired wound healing in OP and BM/MM patients at risk of MRONJ.

Length of drug holiday varies in the literature from a few weeks to several months depending on the published protocols (Hasegawa et al., 2019; Khan et al., 2015; Ruggiero et al., 2014).

That said, data on BP drug holidays never proved robust enough in clinical and animal studies to support its routine use as a preventive measure of MRONJ and that drug holiday is not based on data but considering the mechanism of action and kinetics of the drug (Hasegawa et al., 2021; Jung et al., 2019; Kang et al., 2020; Ottesen et al., 2020).

The SIPMO-SICMF Expert panel agrees on the lack of evidence supporting the scientific validity of prophylactic BP discontinuation (prophylactic drug holiday) prior to oral surgical procedures including tooth extraction.

Instead, the SIPMO-SICMF Expert panel considers a targeted (short-term) interruption of BPs mainly in HD-BP R_+ , HD-BP R_{++} and LD-BP R_x patients a reasonable strategy with limited side effects to prevent the possible alveolar bone accumulation of BP following oral surgical procedures, related to bone remodelling process and

minimize toxicity to the oral mucosa (Table 6) (Hadaya et al., 2021; Inoue et al., 2021; Kobayashi et al., 2010; Ravosa et al., 2011).

Short-term interruption of BPs should start one week before surgery and last until soft-tissue healing has been achieved (4–6 weeks) (Campisi, Mauceri, et al., 2020).

In any case, discontinuation of BPs must be authorized by the treating physician/prescriber, who should outweigh the risks associated with discontinuation of therapy.

Differently from BPs, DMB has a short half-life, and its effect on bone resorption gradually declines within six months after the last dose (de Oliveira et al., 2016). The reversible mechanism of DMB on bone mineral density (BMD) supports the theoretical concern about a possible increased risk of fracture upon stopping DMB (Prolia®) in OP patients (i.e. rebound phenomenon), which usually happens 8–16 months from the last denosumab injection (Anastasilakis et al., 2017; McClung et al., 2017). Indeed, while denosumab therapy increases BMD and reduces fracture risk, the disruption of bone architecture caused by osteoporosis is not reversed with treatment. In this view, withdrawal of Prolia® is largely discouraged in OP patients for the increased risk of rebound-associated vertebral fractures (RAVFs), and switching to another osteoporosis therapy is recommended (Tsourdi et al., 2021).

In the context of preventing skeletal-related events (SREs), DMB (Xgeva®) is initiated at diagnosis of bone metastases and continued indefinitely in many cases (Planhard et al., 2018).

DMB prophylactic drug holiday has been suggested to prevent delayed bone healing and reduce the risk of MRONJ onset in HD-DMB R_+ and HD-DMB R_{++} , but the results are controversial (Hasegawa et al., 2021; Ottesen et al., 2020).

Within the limits of the available knowledge, the SIPMO-SICMF Expert Panel considers DMB prophylactic drug use before oral surgical procedures hazardous and recommends a different approach to be used in patients at increased MRONJ risk, based on the dosing

TABLE 6 Timing of perioperative withdrawal of different ONJ-related medications (modified from Campisi, Bedogni et al., 2020).



schedule of DMB (Xgeva® 120mg Q4W vs. 60mg Prolia® 60mg twice-yearly). In detail, indicated surgical treatments including tooth extraction (Table 5) should be performed in HD-DMB R₊ and HD-DMB R₊₊ patients at least 3 weeks after the last dose of Xgeva®, and the following injection postponed until soft-tissue healing is achieved (4–6 weeks). Specific MRONJ risk reduction protocols must be combined to further reduce the risk of MRONJ onset (see paragraph on dental management).

On the other side, oral surgical procedures including non-urgent tooth extraction (Table 5) can be performed without restrictions in LD-DMB R_x patients 5 months after the last dose of Prolia®, taking advantage of the reactivated bone turnover and of the recovered healing capacity of bone. This “window of opportunity” lasts about 2 months. A targeted 1-month delay of the scheduled dose of Prolia® may be enough to foster soft-tissue healing, without the added risk of RAVFs (Lyu et al., 2020).

Non-deferrable extraction of compromised teeth should be performed at least 3 weeks after the last dose of Prolia®, and specific MRONJ risk reduction protocols combined to further minimize the risk of MRONJ onset (see paragraph on dental management).

Since the approval of AAs for the treatment of a wide range of cancer types, sporadic cases of MRONJ have been reported, especially when they are coupled with BMAs (Srivastava et al., 2021). AAs possess a short half-life as compared with BMAs. These targeted medications not only exert an antiangiogenic effect on tumour cells but also on healthy tissues, thus reducing the ability of soft tissue to repair and complicate postoperative wound healing (Eming & Krieg, 2006).

The SIPMO-SICMF Expert panel agrees that AAs must be discontinued before any surgical operation, including oral and maxillofacial surgery, and the duration of withdrawal depends on their distinct half-life (Campisi, Bedogni, & Fusco, 2020; Nicolatou-Galitis, Kouri, et al., 2019; Scappaticci et al., 2005). The SIPMO-SICMF Expert Panel affirms that in any case, temporary withdrawal of AAs must be authorized by the treating physician/prescriber, who should outweigh the risks associated with discontinuation of the therapy.

The timing of the perioperative withdrawal of different ONJ-related medications based on their distinct half-life is described in Table 6.

2.11 | Treatment

The management of MRONJ is highly debated. Different treatment protocols for MRONJ have been proposed in the literature that are mainly based on case series and retrospective cohorts. To date, there is no robust evidence to support any specific treatment (Beth-Tasdogan et al., 2022; Moraschini et al., 2021). Similarly, temporary withdrawal of BMAs and AAs following MRONJ diagnosis has been suggested in the past years without conclusive validation (Jacobson et al., 2022; Khan et al., 2015; Lee et al., 2013; Tsourdi et al., 2021).

Standard non-surgical (or medical) therapy, which mainly consists of antiseptic mouth rinses, systemic antimicrobial agents, analgesics and smoothing of exposed and sharp bony edges to prevent ulcerated lesions of the oral mucosa, has been recommended as the mainstay of MRONJ treatment for almost 20 years with the aim of eliminating pain and infection and minimizing disease progress (Ruggiero et al., 2022).

2.11.1 | Antibiotic therapy

The fact that infection accompanies, if not determines, the clinical manifestations of MRONJ in most cases supports the use of antibiotics in managing the related signs and symptoms of MRONJ.

Although several clinical studies have reported the use of antibiotics in the management of overt MRONJ, they were always associated with other medical or surgical interventions and administered with different schedules to heterogeneous patient populations, and in the absence of control groups (Hoefert & Eufinger, 2011; Moretti et al., 2011).

Until now, antibiotic therapy is more empirical than based on evidence, regarding the agents to be used, the route of administration, dosages and the treatment duration.

The most used antibiotics for the treatment of MRONJ in patients taking oral BMAs are penicillin, amoxicillin, amoxicillin/clavulanic acid, metronidazole and/or a combination thereof (Bermúdez-Bejarano et al., 2017).

The SIPMO-SICMF Expert Panel, recognizing the importance of systemic antibiotic therapy in the treatment of MRONJ-related infection and pain, has identified an empirical antimicrobial protocol based on the following principles:

1. Combinations of antibiotics including penicillin (active against Gram-negative and Gram-positive β -lactamase-resistant bacteria) and metronidazole (active against anaerobes, particularly *Bacteroides* spp. and Gram-positive cocci), as the first-line treatment;
2. Duration of therapy ranging from a minimum of 7 days to a maximum of 14 days at full dosage;
3. Oral administration route for non-hospitalized patients; preferential I.V. administration for patients hospitalized due to infectious complications or associated surgical procedures;
4. Alternative agents (e.g. erythromycin, clindamycin or ciprofloxacin) to be used in case of allergies to penicillin/cephalosporins or proven inefficacy of standard treatment.

Additional non-invasive treatments have been proposed as an adjunct to non-surgical and surgical therapy, including hyperbaric oxygen therapy, ozone therapy, low-level laser therapy, teriparatide, pentoxifylline (associated or not with tocopherol), autologous platelet-rich blood derivatives and Bone Morphogenetic Proteins, with varying degree of success in selected patient populations (Cavalcante & Tomasetti, 2020; Del Fabbro et al., 2015; Di Fede et al., 2022; Fortunato et al., 2020;

Freiberger et al., 2012; Morishita et al., 2020; Ripamonti et al., 2012; Vescovi et al., 2014).

Yet, these adjunctive treatments require definitive confirmation before being endorsed in routine clinical practice, due to extra costs and restricted availability (Beth-Tasdogan et al., 2022; Di Fede et al., 2021).

A symptomatic treatment approach as described above was based on the assumption that MRONJ is incurable and MRONJ occurs most frequently in metastatic cancer patients with comorbid conditions and limited life expectancy. Consequently, surgical therapy had been considered for a long time palliative rather than curative and offered only to patients with an advanced disease not responding to medical treatment (Advisory Task Force on Bisphosphonate-Related Osteonecrosis of the Jaws, 2007; El-Rabbany et al., 2017; Ferlito et al., 2012; Khan et al., 2015; Ruggiero et al., 2009, 2014; Saussez et al., 2009; Scoletta et al., 2010; Van den Wyngaert et al., 2009; Wutzl et al., 2008).

These statements have been substantially challenged in recent years when it has been cleared out that MRONJ can be cured if properly and timely managed, and that MRONJ patients can greatly benefit from surgery in terms of improved quality of life, restored oral function and access to potentially life-prolonging therapies, including BMAs (Bedogni et al., 2011; Oteri et al., 2017; Ruggiero et al., 2022).

The aim of MRONJ treatment has recently shifted from palliative to curative, thanks to the reliable results of surgery (Bettini et al., 2021; Carlson, 2014; Ruggiero & Kohn, 2015; Schiodt et al., 2019).

In fact, it is now evident that surgical treatment, in combination with medical therapy, offers more predictable results than non-surgical therapy alone in all disease stages and in the long term (Di Fede et al., 2021; Fliefel et al., 2015; Kim et al., 2017).

The SIPMO-SICMF Expert Panel has long been considering surgery the backbone of MRONJ treatment, while being aware that medical therapy still represents a reasonable treatment option for patients unfit for surgery or who refuse it (Bedogni et al., 2013; Campisi et al., 2014; Campisi, Bedogni, & Fusco, 2020; Campisi, Mauceri, et al., 2020). Decision on surgical versus non-surgical treatment remains patient-specific and should always undergo careful clinical judgment.

2.11.2 | Medication withdrawal in confirmed MRONJ cases (therapeutic drug holiday)

The benefits of antiresorptive therapy far outweigh the risk of MRONJ onset. Nevertheless, when MRONJ is diagnosed, BMAs are often discontinued until disease resolution is achieved in many cases (Hayashida et al., 2020; Morishita et al., 2022).

The term “*Therapeutic Drug Holiday*” is used to define the withdrawal of BMAs and AAs soon after the occurrence of MRONJ, with the aim of promoting/accelerating disease resolution. Conceptually, it highly differs from “*prophylactic drug holiday*” that is aimed at

preventing MRONJ in at-risk patients, before oral surgical procedures take place.

It is controversial as to whether therapeutic drug holiday in combination with medical and/or surgical treatments might promote or accelerate MRONJ disease resolution.

Early clinical recommendations and international consensus statements theorized the withholding of BMAs, especially in the oncological setting (Khan et al., 2015; Lee et al., 2013). More recent publications raised concern about the real benefit and potential harm of temporary BMA discontinuation in cancer and OP patients with suspected or established MRONJ and left the final decision only at the discretion of the treating physician, after discussion with the patient and the oral health provider (Coleman et al., 2020; Jacobson et al., 2022; Tsourdi et al., 2021; Yarom et al., 2019).

Based on the different pharmacodynamics and kinetics of BPs as compared with DMB and AAs, the SIPMO-SICMF Expert Panel decided to address separately the likelihood of temporary medication withdrawal in BP-associated ONJ and non-BP-associated ONJ (DMB and AAs).

Despite the lack of robust data on the effectiveness of drug discontinuation, there is general agreement among maxillofacial surgeons and oral health care providers that long-term withdrawal of BPs does not influence the natural course of MRONJ and does not increase the chance of better treatment outcomes (Hayashida et al., 2020; Morishita et al., 2022; Omori et al., 2022; Otsuru et al., 2022).

In light of the long-lasting inhibitory effect of BPs on bone remodelling and in the absence of exhaustive data, the SIPMO-SICMF Expert Panel considers BP withdrawal a potentially harmful strategy for BRONJ patients and does not recommend its routine application, at least in the first years of BP treatment. Alternatively, perioperative discontinuation of BPs (short-term interruption) starting the week before surgery until complete healing is obtained seems a reasonable strategy with limited side effects to reduce their accumulation in the surgical site, although evidence remains weak.

Temporary withdrawal of BPs should be limited to specific cases, decided by the treating physician in agreement with the oral health provider and the pros and cons discussed with the patient.

Since its approval for the prevention of SREs in metastatic cancer patients and the prevention of fragility fractures in OP patients, DMB has been associated with the occurrence of MRONJ at least the same as the most potent BPs (i.e. monthly zoledronic acid and oral BPs in the two population subgroups). As previously mentioned, DMB does not bind to hydroxyapatite and incorporate into bone; thus, bone turnover recovers rapidly after drug discontinuation.

In light of the supposed “rebound effect” phenomenon and the increased risk of RAVFs in OP patients (see paragraph on prophylactic drug holiday), DMB withdrawal is generally contraindicated in LD-DMB patients (Cummings et al., 2018). On the contrary, the reported risk of RAVFs after DMB discontinuation in metastatic cancer patients appears low as compared with OP patients, with only a few reports in the literature (Dupont et al., 2022). It is, therefore,



likely that DMB treatment might be temporarily interrupted in patients with a favourable prognosis who receive HD-DMB, and resumed in case of disease progression (Coleman et al., 2020; Dupont et al., 2022).

Data on DMB withdrawal are scanty but it seems that discontinuation before surgery may accelerate MRONJ resolution (Hayashida et al., 2020; Omori et al., 2022; Otsuru et al., 2022).

Within the limits of the available knowledge, the SIPMO-SICMF Expert Panel considers a 6-month temporary DMB interruption a potentially suitable strategy to reduce the burden of surgical treatment for BP-naïve metastatic cancer patients with MRONJ. On the contrary, DMB withdrawal is not recommended as an adjunctive non-surgical treatment.

Temporary withdrawal of DMB requires accurate clinical judgment and should always be decided on a per-patient basis by the treating physician in agreement with the oral health provider and the pros and cons discussed with the patient. Close clinical and instrumental monitoring of bone turnover is required for the patient's safety and the prompt resumption of BMAs when needed.

In the case of OP and CTIBL patients with MRONJ receiving LD-DMB therapy, the SIPMO-SICMF Expert Panel hypothesizes that a planned 1-month delay of the scheduled dose of DMB may be sufficient to allow surgical operations of reduced intensity to be done 5 months after last DMB injection, while ensuring the benefit of the reactivated bone turnover on the healing capacity of bone, without the added risk of RAVFs.

As previously stated (see paragraph on Prophylactic drug holiday), AAs should be always interrupted before any surgical operation takes place, and the timing of interruption depends on the half-life of the given drug. AA withdrawal should last until complete soft-tissue healing has been achieved (4–6 weeks).

2.11.3 | Surgical treatment

The SIPMO-SICMF Expert Panel recognizes the existence of several points at issue that could negatively influence the result of surgery, including: *a-* the adoption of a single surgical algorithm to treat all MRONJ patients, regardless of their underlying condition (BM/MM vs. OP), the medication received (BPs vs. DMB vs. AAs) and its dosage (high-dose vs. low-dose); *b-* the adoption of a stage-related treatment algorithm based on clinical signs and symptoms only; *c-* the lack of a common strategy to define the surgical bone margins; *d-* the ambiguity of operational definitions proposed for the different surgical interventions; *e-* the absence of standardized outcomes of surgery and the variability of follow-up.

1. There is a growing body of literature that BM/MM patients receiving high-dose BMAs, besides an increased risk of MRONJ, also display earlier disease onset, faster progression/severity and worse prognosis as compared with OP patients receiving low-dose BMAs (Hayashida et al., 2017).

In addition, while BPs have high skeletal retention and are stored in bone for a long time, DMB does not incorporate into the bone, and its inhibitory effect on bone turnover reverses at the end of the dosing interval (McClung et al., 2006). Likewise, the antiangiogenic activity of biological agents is time-dependent, and the blood supply to the bone and surrounding soft tissues goes back to normal a few days/weeks after drug cessation (Fusco et al., 2016; Nicolatou-Galitis, Kouri, et al., 2019).

Overall, these facts make it very likely that *response to surgery* is *not uniform* in MRONJ patients and varies with the patient clinical condition, the medication received, the dosing schedule and intervals. In other words, a given surgical procedure may be excessive for some patient categories (i.e. overtreatment) or deficient for others (i.e. undertreatment).

The SIPMO-SICMF Expert Panel, unlike other groups of experts, endorsed this hypothesis and in 2013 developed a surgical algorithm where the magnitude of surgical therapy was graded among BM/MM patients receiving high-dose BPs and OP patients receiving low-dose BPs (Bedogni et al., 2013; Campisi et al., 2014). Clinical results seem to confirm the need for reduced surgical treatment intensity in MRONJ patients receiving LD-BMAs (Bettini et al., 2021).

2. AAOMS task force was the first to introduce a staging system based on clinical signs and symptoms to address disease severity, cluster patients and assign treatments accordingly (Advisory Task Force on Bisphosphonate-Related Osteonecrosis of the Jaws, 2007; Ruggiero et al., 2009, 2014).

This is still the most largely recognized classification method, despite its failure to depict the real extent of osteonecrosis (Baba et al., 2018; Bedogni et al., 2014; Kim et al., 2020). Therapies that rely on the use of a staging system based on clinical signs and symptoms may prove unsafe now that AAOMS has opened to the surgical treatment of MRONJ for all AAOMS disease stages (Ruggiero et al., 2022).

The SIPMO-SICMF clinical-radiological staging system was developed in 2012 to provide an accurate description of disease extent as the basis for the appropriate delivery of surgical treatments of increasing intensity (Bedogni et al., 2013; Bedogni, Fusco, et al., 2012). The identification of increased bone density (i.e. osteosclerosis) at CT was established as a marker of disease severity to identify increasing levels of bone involvement, ranging from focal to diffuse disease. Surgery is graded so that patients with “focal disease” (Stage 1) are likely to receive less invasive surgical treatment (i.e. bone curettage and sequestrectomy) as compared with more advanced disease stages (Stage 2 and 3), who deserve more radical interventions (i.e. marginal and segmental resection of bone) (Campisi et al., 2014).

The recently proposed AAOMS surgical algorithm parallels the one originally developed by the SIPMO-SICMF Expert Panel,

TABLE 7 MRONJ Treatment.

(a) Stage-related surgical algorithm of BRONJ ^a		
SIPMO-SICMF staging system	High-dose BPs	Low-dose BPs
Stage 1 (focal ONJ)	<i>Dentoalveolar surgery: marginal resection</i>	<i>Dentoalveolar surgery: curettage and bone sequestrectomy; marginal resection for recurrent disease</i>
Stage 2 (diffuse ONJ)	<i>Segmental resection + bone reconstruction when indicated</i>	<i>Dentoalveolar surgery: marginal resection; segmental resection for recurrent disease</i>
Stage 3 (complicated ONJ)	<i>Segmental resection + bone reconstruction when indicated</i>	<i>Segmental resection + bone reconstruction when indicated</i>
• 1-month postoperative BP withdrawal starting 1 week before surgery to reduce accumulation at the surgical site that could hamper the healing process. • Dento-alveolar surgery can be done under loco-regional anaesthesia. • Achieve stable mucosal coverage of the operated site irrespective of the surgical technique adopted. • Adjunctive treatment options: use of piezoelectric or laser-assisted surgery to minimize ischemic damage to the bone. • Perioperative topical disinfection (chlorhexidine 0.2%) for 7–10 days. • Peroral antibiotic therapy (7–14 days long) for surgery under loco-regional anaesthesia; I.V. perioperative antibiotic therapy (7–14 days long) for hospitalized patients and surgery under general anaesthesia. • Perioperative pain control • Postoperative clinical follow-up at 1, 3, 6 and 12 months. CT scans at 6 and 12 months after surgery.		
(b) Stage-related surgical algorithm of DMB-ONJ ^b		
SIPMO-SICMF staging system	High-dose DMB	Low-dose DMB
Stage 1 (focal ONJ)	<i>Dentoalveolar surgery: curettage and bone sequestrectomy; marginal resection for recurrent disease</i>	<i>Dentoalveolar surgery: curettage and bone sequestrectomy; marginal resection for recurrent disease</i>
Stage 2 (diffuse ONJ)	<i>Dentoalveolar surgery: curettage and bone sequestrectomy; marginal resection for recurrent disease</i>	<i>Dentoalveolar surgery: curettage and bone sequestrectomy; marginal resection for recurrent disease</i>
stage 3 (complicated ONJ)	<i>Dentoalveolar surgery: marginal resection; segmental resection for recurrent disease</i>	<i>Dentoalveolar surgery: marginal resection; segmental resection for recurrent disease</i>
• In patients receiving Xgeva®, 6-month temporary DMB interruption is required before surgery. • In patients receiving Prolia®, surgery should be performed 5 months after the last dose, with 1-month postoperative delay of the following dose, until complete healing is achieved. • Dento-alveolar surgery can be done with loco-regional anaesthesia. • Achieve stable mucosal coverage of the operated site irrespective of the surgical technique adopted. • Adjunctive treatment options: use of piezoelectric or laser-assisted surgery to minimize ischemic damage to the bone. • Perioperative topical disinfection (chlorhexidine 0.2%) for 7–10 days. • Peroral antibiotic therapy (7–14 days long) for surgery under loco-regional anaesthesia; I.V. perioperative antibiotic therapy (7–14 days long) for hospitalized patients and surgery under general anaesthesia. • Perioperative pain control. • Postoperative clinical follow-up at 1, 3, 6 and 12 months. CT scans at 6 and 12 months after surgery.		
(c) SIPMO-SICMF Non-surgical treatment protocol of MRONJ		
SIPMO-SICMF staging system	Medical therapy ^c	
All disease stages	• Oral mouth rinses • Systemic broad-spectrum antibiotics (7–14-day long in case of recurrent pain and suppuration; monthly schedule in refractory cases) • Pain control • Adjunctive treatment options: biostimulation with ozone or low-level laser therapy/laser photobiomodulation Aims: symptomatic (palliation); Exfoliation of the exposed, necrotic bone	

^aIt also includes MRONJ patients who have been shifted from NBPs to DMB.

^bSuitable for BP-naïve patients taking DMB.

^cSuitable for systemically compromised patients for whom surgical therapy is contraindicated, or in case of patient's refusal of surgery.



except that they do not detail how to measure the extent of bone disease (i.e. focal vs. diffuse bone involvement).

Being concerned that delivery of surgical treatments of increasing magnitude based on clinical signs and symptoms only will pose MRONJ patients at risk of failure/relapses (undertreatment) or harmful side effects (overtreatment), the SIPMO-SICMF Expert Panel recommends the adoption of its clinical-radiological staging system to assign surgical treatment based on the radiological extent of bone involvement.

3. The amount of bone that needs to be removed surgically is paramount to prevent failures and early MRONJ relapses. Demarcation of necrotic and viable bone margins for surgery is challenging. While several authors agree on the use of preoperative imaging to better visualize the extent of bone disease, many clinicians rely on the intraoperative inspection of bone to define the margins for a "safe" resection (Bedogni et al., 2011; Ruggiero et al., 2022). Both approaches have limitations.

Among the different imaging techniques used, CT and MRI give a detailed description of the bone marrow changes and help define the boundaries of viable and necrotic bone, with respect to the uninvolved bone tissue (Bagan, Leopoldo-Rodado, et al., 2017; Huber et al., 2020).

Multidetector CT (MDCT) and cone beam CT (CBCT) can also pick up the early radiological aspects of MRONJ, as compared with plain radiographs (Devlin et al., 2018; Guo et al., 2016).

Although no distinct imaging phenotype exists for MRONJ and advanced imaging modalities might overestimate the extent of disease as compared with the clinical picture, CT and CBCT could be used to accurately diagnose the extent of MRONJ lesions (Kim et al., 2020).

Establishing healthy bone margins intraoperatively remains a major challenge. Surgeons must observe the bone colour and bleeding to distinguish the necrotic from the viable bone. Bone bleeding has been used for a long as a marker of bone health, but it proved unreliable and operator-biased (Ristow et al., 2016).

Fluorescence-guided bone surgery, with and without tetracycline fluorescence labelling, has been successfully used to increase the intraoperative accuracy of bone surgery (Ristow et al., 2016). The fluorescence technique likely improves the demarcation of viable bone, as compared with clinical inspection alone (Giovannacci et al., 2019; Ristow et al., 2016; Ristow & Pautke, 2014). Since the way for surgical management of MRONJ has been recently cleared, the SIPMO-SICMF Expert Panel is concerned that the absence of a common strategy to define the surgical bone margins of MRONJ will negatively

impact the results of future clinical research on this topic and preclude comparison of different surgical treatment strategies.

Because MRONJ is a bony disease, the SIPMO-SICMF Expert Panel recommends the adoption of CT-based imaging techniques to preoperatively assess MRONJ patients elected for surgery, for two main reasons: 1- the same imaging technique can be used to diagnose, stage and define the extent of the disease; and, 2- identification of the margins of bone surgery before the operation takes place enables the effective exchange of information between physicians and patients concerning the proposed treatment and its consequences.

Also, the SIPMO-SICMF Expert Panel supports the use of MR imaging techniques in combination with CT-based imaging to improve definition of the margins of bone resection in selected cases.

4. Surgical procedures generally used to treat MRONJ patients range from superficial bone debridement to more radical interventions including the whole resection of diseased bone, with or without simultaneous reconstruction.

The lack of standardized terminology to label MRONJ surgical procedures of different magnitude and anatomical location has made it difficult to compare treatment results in the past years. Different surgical procedures have been confused under the same term or as opposed the same surgical procedure has been termed differently (Schiodt et al., 2019).

The SIPMO-SICMF Expert Panel endorses the adoption of a standardized set of surgical interventions to establish the risks and benefits of each operation and allow disease stage-related comparisons of different treatments.

5. Additional limitations of the current approach to MRONJ treatment are the lack of standardized outcomes to assess the impact of surgical procedures on patients, and the high variability of postoperative follow-up (Choi et al., 2020).

Although the curative potential of surgery signifies a paradigm shift in MRONJ treatment compared to the past, little has been done to refine the criteria of treatment success adopted so far. In this view, the widely used definition of painless mucosal healing seems insufficient as it only reflects the condition of the covering mucosal surface (Beth-Tasdogan et al., 2022).

The SIPMO-SICMF Expert Panel suggests the adoption of a more comprehensive definition of healing that encompasses the condition of the underlying bone. This can be achieved by coupling the clinical and radiological criteria of healing (Campisi, Mauceri, et al., 2020) so that treatment can be considered successful, and healing completed in the absence of both clinical and radiological signs of MRONJ (Campisi et al., 2014); in addition, any given

treatment could also be considered successful in the presence of stable radiological signs of MRONJ when the clinical signs and symptoms are absent (i.e. remission).

The transition from a higher to a lower disease stage (i.e. stage improvement) following treatment of MRONJ has been considered a positive outcome in many published studies in the past years (Comas-Calonge et al., 2017; Ruggiero & Kohn, 2015; Tenore et al., 2020; Vescovi et al., 2006). A combination of "stage improvement" and "mucosal healing" used to generate composite treatment outcomes has raised the success rate of interventions in several MRONJ cohorts but did not improve the clinical condition of treated patients, unless in early-stage diseases (Giudice et al., 2020).

The SIPMO-SICMF Expert Panel is concerned that the inclusion of "stage improvement" in the clinical outcome of MRONJ treatments does not reflect a real advantage for patients as they might experience further disease relapse.

How long surgically treated patients should be followed up before being considered cured is another debated aspect. Published clinical studies on MRONJ treatment are largely retrospective, display limited follow-up times and do not report a precise follow-up schedule for operated patients (Aljohani et al., 2019; Carlson & Basile, 2009; Okuyama et al., 2021; Stockmann et al., 2010). Based on the data available, we can only imply that MRONJ recurrences usually occur within 6 months after surgery, with a considerable number of relapses happening up to 1 year, independent from the type of treatment, the primary disease and the medication received (Bedogni et al., 2011; Choi et al., 2020). Therefore, the optimistic results of surgical therapy described in previous studies with limited follow-up may have been partly overestimated.

The SIPMO-SICMF Expert Panel considers successful any MRONJ treatment (medical and/or surgical) that displays clinical-radiological signs of healing or remission at the 1-year follow-up and suggests the use of such a composite clinical outcome to assess the efficacy of surgical treatments in future clinical studies.

2.11.4 | SIPMO-SICMF stage-related surgical algorithm

The SIPMO-SICMF stage-related surgical algorithm was originally developed in 2013 to specifically address increasing levels of surgical intensity based on the extent of BRONJ and the primary disease of affected patients (Campisi et al., 2014). This algorithm has been recently upgraded to include a separate treatment protocol for DMB-ONJ (Campisi, Bedogni, & Fusco, 2020). The SIPMO-SICMF stage-related surgical algorithm describes the combination of medical therapies and surgical techniques to be adopted in BRONJ (Table 7a) and DMB-ONJ (Table 7b); it also describes the non-surgical treatment of MRONJ (Table 7c). For a detailed description of the SIPMO-SICMF stage-related surgical algorithm including jawbone reconstructive options refer to supplement (Appendix S3, Table IIIa,b).

3 | CONCLUSION AND FUTURE RESEARCH

This Position paper recognizes the weakness of the available evidence on MRONJ and the difficult task that both researchers and clinicians are facing to develop common strategies for the prevention and treatment of this multifaceted disease. The SIPMO-SICMF Expert Panel urges moving towards a universally accepted and less strict definition of MRONJ, an individualized MRONJ risk assessment and the recognition of the essential role of imaging in the diagnosis, classification and management of overt MRONJ.

The SIPMO-SICMF Expert Panel recognizes that the statements and recommendations here provided warrant further confirmation and updates, and highlights the need for a global and interdisciplinary scientific approach to MRONJ to overcome region-specific challenges.

Additionally, the SIPMO-SICMF Expert Panel underlines the importance of communicating the individual risk of MRONJ to patients and caregivers, to warrant patients' adherence to medical treatment and oral health programmes in the long term; it also encourages healthcare professionals to constantly update the literature to be guided in clinical decision-making since new medications with a potential threat to patients come to market; and finally it promotes the large-scale dissemination of the present document among the healthcare professionals involved.

AUTHOR CONTRIBUTIONS

Rodolfo Mauceri: Investigation; data curation; writing – original draft; writing – review and editing; visualization. **Alberto Bedogni:** Conceptualization; methodology; writing – original draft; writing – review and editing; investigation; data curation; supervision; visualization; project administration; validation. **Vittorio Fusco:** Conceptualization; investigation; writing – original draft; writing – review and editing; methodology; validation; visualization; project administration; data curation; supervision. **Francesco Bertoldo:** Validation. **Giordana Bettini:** Validation. **Olga Di Fede:** Validation. **Antonio Lo Casto:** Validation. **Claudio Marchetti:** Validation. **Vera Panzarella:** Validation. **Giorgia Saia:** Validation. **Paolo Vescovi:** Validation. **Giuseppina Campisi:** Conceptualization; investigation; funding acquisition; writing – original draft; methodology; validation; writing – review and editing; project administration; data curation; supervision; visualization.

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CONFLICT OF INTEREST STATEMENT

The authors declare no competing interests. Within the last 12 months, G.C. and A.B. received honoraria for lecturing by Amgen; and F.B. has been consultant or speaker for Abiogen, Amgen, UCB pharma, Chiesi and SPA.

DATA AVAILABILITY STATEMENT

The data that supports the findings of this study are available in the supplementary material of this article.

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SUPPORTING INFORMATION

Additional supporting information can be found online in the Supporting Information section at the end of this article.

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